

Prediction of Chronic Disease Risk in Older
Adults using Nutrition, Lifestyle & Clinical Data
from The Irish Longitudinal Study on Ageing
(TILDA)



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Abstract

Background: Chronic diseases such as Cardiovascular Disease (CVD), Diabetes Mellitus (DM), Frailty (FR), and obesity represent major contributors to morbidity in older adults. While traditional prediction models focus heavily on clinical and functional indicators, the contribution of nutritional and lifestyle factors to early risk detection remains underexplored.

Methods: This thesis applies machine learning techniques to data from The Irish Longitudinal Study on Ageing (TILDA), a national representative longitudinal study of older adults. Five waves of socio-demographic, clinical, nutritional, psychological, and behavioural data, with selected variables from the externally produced Research and Development Corporation (RAND) + Harmonised TILDA dataset, together were systematically cleaned and standardised using a multi-stage preprocessing pipeline. Predictive models, including (*models x , y , z*) were developed to estimate the risk of chronic disease and FR related outcomes. Model performance was evaluated using (*mention examples*), and feature importance was used to assess predictive value.

Results: TBD

Conclusions: TBD

Keywords: ageing, (Artificial Intelligence (AI)), Machine Learning (ML), predictive modelling, Frailty (FR), chronic disease, nutrition, The Irish Longitudinal Study on Ageing (TILDA)

Declaration

I Abderahman Haouit declare that I have produced this thesis without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This thesis has not previously been presented in identical or similar form to any other Irish or foreign examination board.

The thesis work was conducted from 2025 to 2026 under the supervision of Dr. Alison O'Connor at University of Limerick.

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Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

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List of Abbreviations

AI	Artificial Intelligence
AUC	Area Under the ROC Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
BMI	Body Mass Index
BP	Blood Pressure from Sphygmomanometer
brainPAD	Brain predicted age difference
CAPI	Computer Assisted Personal Interview
CES-D	Center for Epidemiologic Studies Depression Scale
COG	Scales and Questions on Cognition
CRT	Choice Reaction Time (Cognitive Test)
CV	Cross Validation
CVD	Cardiovascular Disease
DIS	Disability, Functional Impairment and Helpers
DM	Diabetes Mellitus
DNN	Deep Neural Network
EU	European Union
F1	F1-score (harmonic mean of precision and recall)
FOF	Fear of Falling
FR	Frailty
GBM	Gradient Boosting Machine
GDPR	General Data Protection Regulation
HADS	Hospital Anxiety and Depression Scale
HGS	Hand Grip Strength

List of Abbreviations

IADL	Instrumental Activities of Daily Living
INC	Income Variables
IPAQ	International Physical Activity Questionnaire
ISSDA	Irish Social Science Data Archive
JSON	JavaScript Object Notation
MD	Medication Data
MGS	Maximum Gait Speed
MH	Scales and Questions on Mental Health and Wellbeing
ML	Machine Learning
MVPA	Moderate-to-Vigorous Physical Activity
NOS	Newcastle-Ottawa Scale
OR	Odds Ratio
PA	Physical Activity
PMF	Pseudonymised Microdata File
RAND	Research and Development Corporation
Recall	Recall (true positive rate)
RF	Random Forest
SCQ	Self Completion Questionnaire
SCR	Screening Tests
SHAP	Shapley Additive Explanations
SLR	Systematic Literature Review
SOC	Social Variables
SPSS	Statistical Package for the Social Sciences
SR	Systematic Review
TILDA	The Irish Longitudinal Study on Ageing
TUG	Timed Up and Go (mobility test)
UGS	Usual Gait Speed
WHO	World Health Organization

Chapter 1

Introduction

Ageing populations experience substantial changes across physical, nutritional, cognitive, and psychosocial domains, each shaping vulnerability to chronic disease and functional decline [6–8]. Understanding how these multidomain factors interact over time is essential for early risk identification.

The financial and societal burden is substantial. In Europe, poor diet is a major risk factor for non communicable diseases, contributing billions in health-care costs annually [9]. In Ireland, malnutrition alone is estimated to cost over €1.4 billion per year [10]. In response to these challenges, international public health frameworks, including the World Health Organization (WHO) *Global Action Plan* on noncommunicable diseases and Ireland’s *Healthy Ireland* framework, highlight nutrition and lifestyle factors as central components of disease prevention strategies.

European regulatory frameworks such as the General Data Protection Regulation (GDPR) and proposed European Union (EU) Artificial Intelligence (AI) Act emphasise transparency and highlight the need for interpretable Machine Learning (ML) models in healthcare.

Traditional risk prediction models have largely relied on clinical biomarkers such as Blood Pressure from Sphygmomanometer (BP) and lipid profiles. Yet, growing evidence shows that broader predictors, including nutrition [11], Physical Activity (PA) [12], Scales and Questions on Mental Health and Wellbeing (MH) [13], and Social Variables (SOC) [14] improve predictive performance. Es-

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timating the likelihood of developing a condition given certain risk factors helps quantify the influence of these predictors in clinical decision making.

TILDA¹ provides a national representative cohort of adults aged 50+ in Ireland, surveyed repeatedly across multiple domains. For cross study comparability and methodological consistency in model development, selected variables from the externally produced RAND Harmonised TILDA dataset (available for Waves 1 and 2) were included.

The harmonised dataset provides standardised variable definitions and documented construction rules, which are particularly relevant in a machine learning setting where models are trained and evaluated on partitioned data. Its inclusion supports robust feature engineering, reproducible train validation splits, and alignment with prior ageing related ML studies that adopt harmonised frameworks. The specific role of harmonised variables within the modelling pipeline is detailed in Chapter 3.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over reliance on clinical predictors, limited incorporation of lifestyle and SOC variables, predominantly cross sectional study designs that fail to capture trajectories, and a lack of interpretability consistent with emerging AI governance standards.

This thesis addresses these gaps by leveraging longitudinal TILDA data across multiple domains, using interpretable ML methods to combine modifiable health behaviours, psychosocial factors, and established clinical measures to enhance prediction of ageing related outcomes.

1.1 Research problem and motivation

Despite increasing research and clinical interest in lifestyle, nutrition factors and advances in ML methods, approaches to chronic disease prediction in older adults remain limited in several ways.

¹TILDA's dataset are pseudonymised and accessed under license via ISSDA. Use adheres to UL S&E ethics approval and data retention requirements.

- **Limited Integration of Multidomain Predictors:** Existing models focus primarily on clinical measurements and often neglect modifiable behavioural, dietary, and psychosocial factors [11, 15].
- **Underuse of Longitudinal Dynamics:** Few studies leverage repeated measures to capture developments of risk, such as FR progression [16]. Probabilistic reasoning can help estimate the likelihood of developing a condition at a future time given earlier measurements, for example, calculating the probability $P(\text{FR at wave 5} \mid \text{UGS at wave 1, Nutrition at wave 2})$.

This approach helps quantify how early indicators influence later outcomes and supports better feature selection and timing of interventions in predictive models.

- **Lack of Explainability:** Many ML models are not interpretable, limiting clinical adoption and failing to meet emerging regulatory requirements for trustworthiness AI in healthcare [17].

Addressing these challenges is essential to improve predictive accuracy, support evidence based interventions, and ensure alignment with evolving policy and regulatory priorities.

1.2 Research aims and objectives

The goal of this thesis is to develop and validate ML models that integrate nutrition, lifestyle, and MH features with clinical data to improve prediction of chronic disease and ageing related outcomes in a representative cohort of older adults. This goal will be achieved through the following objectives:

- Review existing applications in ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.
- Develop a reproducible pipeline to preprocess and align TILDA waves 1–5, and engineer multidomain features (nutrition, activity, mobility, MH, clinical indicators).

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- Develop and compare baseline statistical models (e.g., logistic regression) with advanced interpretable ML methods such as Random Forest (RF) and Gradient Boosting Machine (GBM).
- Evaluate models using predictive classification performance metrics including Area Under the ROC Curve (AUC), calibration and error measures (accuracy, precision, Recall (true positive rate) (Recall), F1-score (harmonic mean of precision and recall) (F1)), perform ablation analyses to assess the incremental value of lifestyle and nutrition predictors.
- Identify actionable predictors (e.g., vitamin D status, UGS reserve) and highlight modifiable risk factors for intervention.
- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.3 Contributions of the research

This research makes contributions at three levels:

- **Empirical:** Demonstrates how behavioural, dietary, and other MH features complement established clinical measures to enhance predictive performance.
- **Methodological:** Provides a reproducible pipeline for aligning longitudinal TILDA data across domains and integrating diverse features into ML workflows.
- **Clinical:** Highlights actionable modifiable factors alongside traditional clinical indicators for targeted interventions and policy.

1.4 Thesis Outline

In *Chapter 2* a literature review on ML prediction of chronic disease in ageing populations, with emphasis on lifestyle, nutrition predictors and methodological

considerations, is presented. *Chapter 3* discusses the TILDA study design, data preprocessing, alignment, and feature engineering. In *Chapter 4* the modelling framework is outlined. This includes baseline and advanced ML, hyperparameter tuning, validation, and evaluation metrics. *Chapter 5* shows the model result, including model comparisons, assessments of feature importance, and an ablation analysis. In *Chapter 6* the implications, limitations, and ethical considerations are discussed, and future directions for the research are proposed. Additional supporting documents are included in appendix A.

Chapter 2

Literature Review

This chapter reviews literature on ageing, chronic disease, and predictive modelling using TILDA (introduced in Chapter 1). It focuses on three interrelated domains central to later life health: FR, physical function, MH, and nutrition within the context of biological ageing. This review focuses on FR, MH, and nutrition as key determinants of later life outcomes. In line with the WHO healthy ageing framework, research using TILDA highlights FR, MH, and nutrition as interconnected determinants of later life outcomes. The following subsections examine each domain in turn.

2.1 Methodology

Several sources describe how systematic and integrative reviews are commonly carried out in fields such as software engineering, health, and education. The approach followed key principles for systematic mapping, for example, defining research questions, identifying relevant studies through structured database searches, applying inclusion and exclusion criteria, and categorising results as outlined in [1]. The integrative literature review [18] helped structure and connect findings across studies, while [19, 20] guided the organisation of ideas. In related methodological work, [2] clarified how SRs can identify research gaps, and [21] showed how mapping and synthesis highlight trends and future directions. This combined guidance informed a structured yet flexible methodology suitable for ageing in AI, and health research.

2.1.1 Search strategy and database screening

The main topic of interest was chronic disease prediction using nutrition, lifestyle, and clinical data from TILDA. Relevant databases (e.g., ScienceDirect, PubMed) were searched using combinations of keywords such as “Irish Longitudinal Study on Ageing”, “machine learning”, “chronic disease”, “nutrition”, “diet”, and “physical activity” (see table 2.1).

The search covered studies published from 2011 to 2025. Figure 2.1 summarises the number of studies identified per year, highlighting trends and gaps over time.

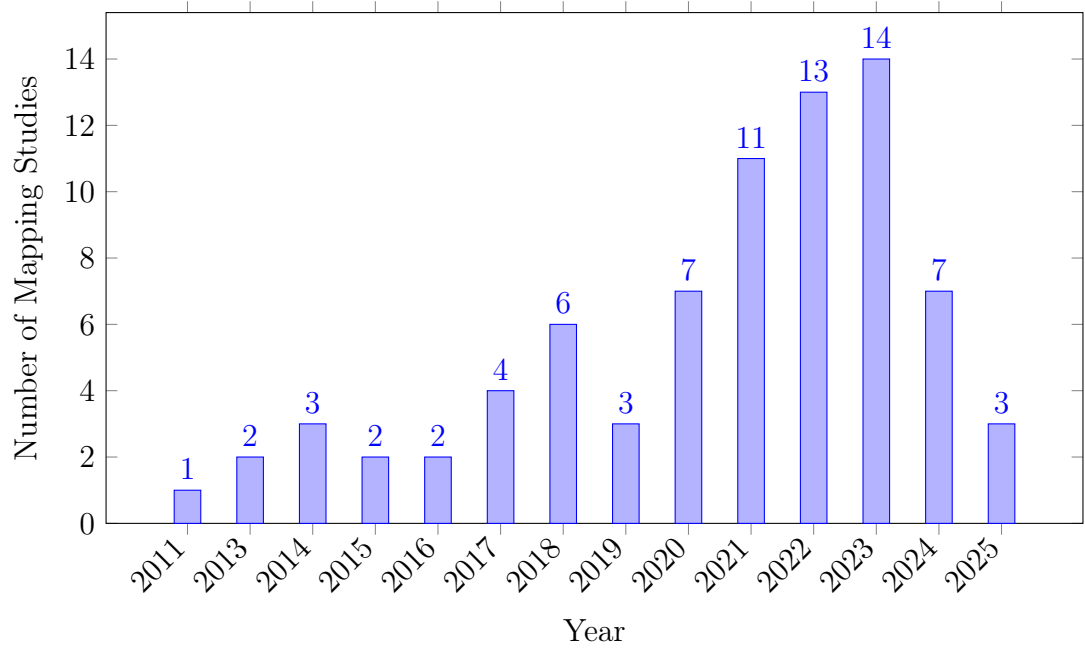


Figure 2.1: Publications per year.

Some search strings returned few or irrelevant results, particularly for ML related TILDA studies, highlighting a key research gap. Iterations continued until suitable combinations consistently returned relevant studies. Duplicates were removed using Zotero’s reference management feature. The remaining studies were individually and manually checked for relevance to “ageing,” “chronic disease,” and ML. Using Zotero’s notes functionality, the initially created table 2.3 captured key elements for each paper, including model type, features used, and

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Table 2.1: Database searches and number of studies per database 2.3 [1].

Database	Search	Results
Oxford Academic	(TILDA OR Irish Longitudinal Study on Ageing)AND (chronic disease OR diabetes OR cardiovascular) AND (nu- trition OR diet OR "physical activity")	1081
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk predic- tion")	71
IEEE Xplore	TILDA OR "Irish Longitudinal Study on Ageing"	56
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk predic- tion")	7
ScienceDirect	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR "physical activity")	257
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model") AND ("chronic disease" OR diabetes) AND (nutrition OR "physical activ- ity")	20
PubMed	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR "physical activity")	27
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk predic- tion")	9

main findings for each paper in the collection, as illustrated in the PRISMA flow diagram (Figure 2.2).

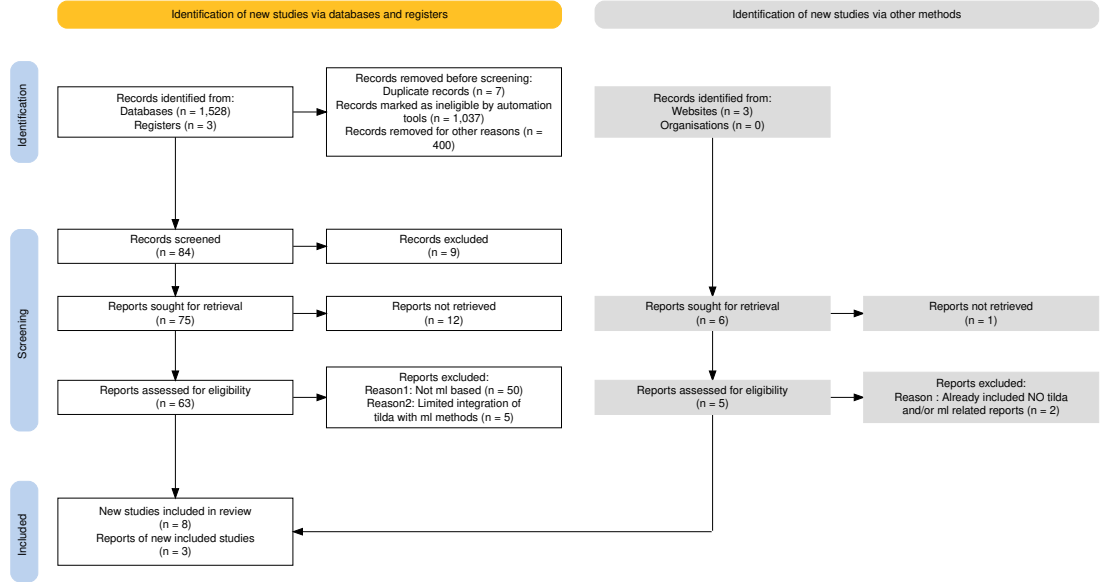


Figure 2.2: PRISMA chart for SRs showing the study selection process for the systematic mapping and integrative review [5].

2.1.2 Quality and relevance evaluation

To systematically assess the quality of the included studies, a customisable version Newcastle-Ottawa Scale (NOS) was applied; see table 2.2. The NOS evaluation studies across three domains: *Selection* (S1–S4), *Comparability* (C1–C2), and *Outcome* (O1–O3). *Selection* criteria assess the representativeness of the sample, clarity of exposure measurement, and temporal alignment of outcomes. *Comparability* examines the control of key confounders (variables that may influence both the predictor and outcome), while *outcome* evaluates measurement validity, follow up adequacy, and completeness. Each domain is scored 0–1, recall the Bernoulli random variable (0 = criterion not met, 1 = criterion met), and the total score reflects the overall study quality.

This initial table guided a further refinement of the results into two sub tables: (1) studies critically related to TILDA, (2) studies using TILDA data ML for

2. LITERATURE REVIEW

predictive contexts, which revealed minimal to no integration with TILDA.

Table 2.2: Quality evaluation of SLRs [2].

Reference	S1	S2	S3	S4	C1	C2	O1	O2	O3	Total
(Xu, 2023 [22])	1	1	1	1	1	1	1	1	1	9
(Donoghue, 2023 [17])	1	1	1	1	1	1	1	0	1	8
(Moguilner, 2021 [16])	1	1	1	1	1	0	1	1	1	8
(Davis, 2021 [6])	1	1	1	1	1	0	1	1	0	7
(Boyle, 2021 [23])	1	1	1	1	1	0	1	0	1	7
(Tzemah-Shahar, 2022 [24])	1	1	1	0	1	0	1	1	1	7
(A. Shirsath, 2024 [25])	1	1	1	1	1	1	1	0	1	8
(Romero-Ortuno, 2021 [26])	1	1	1	1	1	1	1	1	1	9
(Bardon, 2020 [27])	1	1	1	0	1	0	1	1	1	7
(Zhang, 2025 [28])	1	1	1	1	1	1	1	0	1	8
(Sutin, 2023 [29])	1	1	1	1	0	0	1	1	1	7

Table 2.3 summarises selected studies on ageing using TILDA and ML, reflecting key gaps and opportunities identified during systematic mapping. These steps were aligned with the research matrix created at the beginning, ensuring that the refinement of studies and synthesis of findings stayed closely tied to the original research goals.

2.1.3 Synthesis and Integration

Using the NOS scores, each study was annotated in Zotero with details on dataset, key variables, analytical approach, and main findings, facilitating a consistent and systematic evaluation across studies. For the systematic component, data was systematically extracted and synthesised to identify trends (e.g., Markov models, logistic regression, 5-fold Cross Validation (CV)) and research gaps [2], complemented by DoctoralWriting resources [30, 31], which clarified how to define research questions and develop logical arguments.

The integration of a research matrix with systematic mapping and integrative synthesis ensured that the methodology was well suited to capturing the complexity of ageing, nutrition, chronic disease risk and predictive modelling studies focused primarily on TILDA based studies, but due to the limited number of ML

2.1 Methodology

Table 2.3: Summary of selected studies on ageing using TILDA.

Theme	Author/year	Dataset	Variables/features	Methods/model type	Key Findings/outcome	Limitations
FR	Romero-Ortuno (2021) [26]	Wave 1–5	FR state, FR phenotype (FP) components (Exhaustion(EX), Unexplained weight loss(WL), Weakness(WK), Slowness(SL), Low physical activity(LA))	Multi state Markov model, longitudinal 8 year follow up	32% of pre frail to non frail; FR strongly predicted 31% mortality	Self report bias; observational design
Falls + ML	Donoghue (2023) [17]	Wave 1–3	Gait, mobility, medications, history of falls	Conditional inference forest; 5-fold CV; undersampling for imbalance	Accuracy up to 69.7%; for adults aged 50–64.	Area Under the Receiver Operating Characteristic Curve (AUROC) values were ≤ 0.69 and PRAUC ≤ 0.39 indicating low predictive accuracy; no external validation; cross sectional; Class imbalance
Fear of Falling (FOF)	Zarudsky (2014) [32]	Wave 1	Age, sex, fall history	Cross sectional prevalence analysis	23.3% prevalence; higher in women; 70% with FOF had no recent falls in the last 12 months	Self reported; cross sectional
Functional Mobility	Huang (2019) [33]	Wave 1–2	Measured by Timed Up and Go (mobility test) (TUG), Hand Grip Strength (HGS), physical activity, vision, Body Mass Index (BMI), age	Linear regression, stratified analyses	Age modifies effects; PA and grip protect mobility; vision benefit only in middle aged (Beta = -0.032, P = 0.002)	Observational; 2 year follow up
Depression	Jacob (2023) [34]	Wave 1–2	Falls, FOF, HADS-A, Center for Epidemiologic Studies Depression Scale (CES-D)	Prospective logistic regression	Falls predicted incident anxiety Odds Ratio (OR)=1.58 and depression OR=1.43	Self reported falls; short follow up; mediation effect partially explored
	Laird (2023) [15]	Wave 1–5	Moderate-to-Vigorous Physical Activity (MVPA), walking, sociodemographics	Longitudinal regression; mixed model	Higher physical activity reduced depressive symptoms; dose response effect	Self reported PA; observational; limited causal inference
Nutrition	Murphy (2023) [11]	Wave 1–5	Plasma lutein, zeaxanthin	Linear/ordinal logistic regression	Higher carotenoids predicted slower FR progression Lower odds of becoming frail (ORs = 0.57–0.89)	Observational; not predictive of longitudinal muscle changes
	Bardon (2020) [27]	Wave 1–2	Hospitalisation, functional status, SOC support, sex, falls	Logistic regression; sex stratified analyses	Functional decline, hospitalisation, poor support increased malnutrition risk; sex differences	Observational(short follow up (2 years)); limited generalisability >65 years, self reported
ML	Xu (2023) [22]	Wave 1–3	105 predictors from 40,000+ variables (SOC, clinical, mental, family)	Stacked ensemble: RF, XRT, GLM, GBM, Deep Neural Network (DNN); Shapley Additive Explanations (SHAP) explainability	AUC=0.854 for 2 year DM prediction; top predictors LDL, cirrhosis, sex	No external validation; short follow up; ensemble complexity limits deployment
	Davis (2021) [6]	Wave 3	Age, HGS, chair stands, BMI, cognitive tests	Histogram gradient boosting regression; SHAP	r^2 test: UGS $\approx$ 0.41, Maximum Gait Speed (MGS) $\approx$ 0.46, GSR $\approx$ 0.21	Modest predictive power; cross sectional; no external validation
	Zhang (2025) [28]	Longitudinal IMAGEN cohort	Psychopathology, personality, cognition, substance use, environment	Regularised logistic regression; diagnostic	Longitudinal AUC: ED=0.71, MDD=0.64, AUD=0.67; shared transdiagnostic predictors	Limited age range; no external replication
	Boyle (2021) [23]	Wave 3	Brain predicted age difference (brainPAD); processing speed; attention; cognitive flexibility; verbal fluency	Linear regression, correlation analyses	Higher brainPAD associated with lower cognitive performance across multiple domains	Cross sectional; no causal inference; limited sample

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applications within TILDA, additional studies on older adult populations and chronic disease were included.

2.2 Frailty and physical function

FR reflects declining physiological reserve, providing a framework for understanding physical vulnerability in ageing. In TILDA, it is captured through both the FR phenotype and the FR index, with UGS, HGS, and mobility decline serving as practical predictors. These measures strongly predict hospitalisation and mortality.

2.2.1 Probabilistic interpretation of frailty

One can argue that from a probabilistic perspective, the chance of an adverse outcome (such as mortality, Y) given observed features (such as FR markers, X) is a conditional probability $P(Y | X)$. This allows us to deduce which FR features are most predictive of adverse outcomes. Features may be independent, dependent, or mutually exclusive.

The presence or absence of metabolic syndrome represents a mutually exclusive event that can inform risk models.

$$P(X_i \cap X_j) = P(X_i)P(X_j)$$

for independent features X_i and X_j , guiding feature selection and reducing redundancy in model inputs. This reflects the likelihood that an individual with certain FR characteristics will experience a particular outcome.

Evidence suggests that FR is dynamic, not irreversible. For example, multi state modelling showed that many prefrail adults transition back to non frail, although established FR strongly predicts mortality [26]. Metabolic syndrome accelerates the development of FR, suggesting a metabolic pathway of vulnerability [35]. The presence of metabolic syndrome modifies the conditional probability $P(\text{FR} | \text{Metabolic Syndrome})$ and may increase the likelihood of adverse outcomes over time. Another detail is that age matters: obesity predicts mobility decline earlier in life, whereas later in life, traits that protect against this, including HGS, become more significant [33].

These findings partly conflict: FR appears both reversible and progressive, depending on risk context. This complexity highlights the need for probabilistic models that can update beliefs about risk as new features are observed, as formalised by Bayes theorem:

$$P(Y | X) = \frac{P(X | Y)P(Y)}{P(X)}$$

where $P(Y | X)$ is the posterior probability of outcome Y given features X .

2.2.2 Bayesian updating of frailty risk

Causal inference is limited by observational designs, and predictors are frequently examined separately, ignoring interactions. Most FR models have only been with the TILDA dataset, and with this study that focuses on the same TILDA but uses both longitudinal TILDA waves and harmonised variables to improve cross study comparability, enabling for a more robust training validation splits in ML models, and explores richer predictive patterns over time. These gaps point to the need for multidimensional approaches. Traditional regression models may not fully capture age dependent interactions, suggesting a role for more flexible methods such as machine learning techniques and models.

2.3 Mental health in older adults

MH, including depression, anxiety, loneliness, and SOC isolation, strongly shapes quality of life. Evidence from TILDA shows strong links to both physical and SOC factors. [36] found that participation in physical, SOC, and cognitive activities summarised by the Act-Belong-Commit (ABC) indicator was associated with lower depression and anxiety. This suggests SOC integration and lifestyle protect mental wellbeing. However, [37] reported that self reported measures may exaggerate such associations..., raising concerns of bias. These two findings conflict: the effect of SOC engagement is strong when self reported and weaker when objectively assessed.

Objective data strengthen the case for activity. Accelerometer based measures show a dose response link between moderate to vigorous activity and lower

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depressive symptoms [12]. Yet reverse pathways are also evident: incident falls and functional decline predict later depression, refereed by FOF [34]. Together, these results reveal a bidirectional relationship, MH both influences and is influenced by physical decline. Most studies rely on broad self report scales, such as CES-D or Hospital Anxiety and Depression Scale (HADS), with little biomarker or clinical validation.

This literature suggests MH in ageing cannot be modelled as a one way effect. It requires longitudinal, multidimensional approaches that can capture mutual pathways, an area where machine learning is well suited, especially when combined with probabilistic models that estimate the likelihood and posterior probability of MH outcomes given multiple interacting features.

For example, we can estimate:

$$P(\text{Depression} \mid \text{Activity, Falls, SOC Engagement}),$$

quantifying how these interacting features together influence MH outcomes. Probabilistic modelling helps determine which features have the strongest effect and which may be redundant for predictive purposes.

2.4 Nutrition and biological ageing

Nutrition interacts with both biological and psychosocial systems, shaping ageing trajectories and FR. Biomarkers like vitamin D, lutein, and zeaxanthin have been linked to physical and cognitive outcomes in TILDA. [11] showed that higher plasma lutein and zeaxanthin dietary carotenoids found in leafy vegetables were linked to slower FR progression. By contrast, [38] found vitamin D deficiency predicted DM, musculoskeletal decline, especially among adults with low sunlight exposure. These studies point to different nutritional mechanisms: antioxidant protection versus metabolic regulation.

2.4.1 Bayesian perspective on nutritional biomarkers

The effect of nutrition on ageing outcomes can be formalised probabilistically using Bayes theorem:

$$P(\text{Outcome} \mid \text{Nutritional Status}) = \frac{P(\text{Nutritional Status} \mid \text{Outcome})P(\text{Outcome})}{P(\text{Nutritional Status})},$$

where the likelihood function $P(\text{Nutritional Status} \mid \text{Outcome})$ quantifies how observed nutritional biomarkers relate to health states, and the posterior $P(\text{Outcome} \mid \text{Nutritional Status})$ captures updated beliefs about outcomes, including uncertainty in small sample studies. This helps identify which biomarkers are most informative for predicting ageing outcomes which supports feature selection in ML models.

2.4.2 Contextual and integrative considerations

Malnutrition also reflects SOC and functional risks. [27] found that hospitalisation, functional loss, and poor SOC support increased malnutrition risk, with notable sex differences. This highlights a limitation in much nutritional research: biological markers are studied without integrating SOC context.

The difference between biological and chronological age is a major point of debate. Biomarker based metrics that better capture these cumulative exposures, like FR indices or epigenetic clocks, frequently vary from chronological age. McCarthy and Azizi link metabolic syndrome and nutritional deficits to accelerated biological ageing [39, 40]. Although the majority of the evidence is cross sectional and has little ability to capture changes over time, these indicators offer a more precise prediction of vulnerability. Reliability is also decreased in biomarker research with small sample sizes, which can be formalised through the likelihood function $P(\text{Data} \mid \theta)$ and its impact on posterior uncertainty in Bayesian inference.

Taken together, nutrition interacts with biological, metabolic, and SOC systems. Isolating single nutrients risks oversimplification. Integrative modelling that combines biomarkers, clinical factors, and SOC determinants supported by machine learning pipelines offers a more promising route. Models that estimate

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the joint distribution of features and outcomes can better capture the complex dependencies relevant to biological ageing.

2.5 Machine Learning in ageing and chronic disease

ML is increasingly applied in ageing research because it can capture non linear and multidimensional relationships that traditional regression models, such as logistic regression, often miss [22, 23].

2.5.1 Feature sets in ML studies

Studies using ML in ageing research include many types of predictors: clinical measures, sociodemographic factors, lifestyle habits, cognitive tests, and psychosocial variables. A recurring pattern, however, is that models often underline easily measured physical health data, such as comorbidity counts, mobility scores, or vital signs, while more complex but important predictors, including nutrition and SOC factors, are underrepresented [32]. This imbalance may limit the ability of models to capture the full complexity of chronic disease risk with ageing, especially considering that MH and nutrition exert independent and interacting effects on later life outcomes (here). Models that focus only on ready available clinical data risk reproducing the same biases seen in traditional regression approaches.

2.5.2 ML models used

The methodological approaches used in ageing research span both traditional and advanced models. Logistic regression and Markov models remain common baselines due to their interpretability and simplicity, yet they are often unable to capture non linearities or higher order interactions. More recent work has shifted towards ensemble methods such as RF, gradient boosting machines GBM, as well as DNNs, which generally achieve higher predictive accuracy.

Similarly, [41] evaluated mortality prediction using a standard logistic regression model with demographic, clinical, and functional predictors, achieving an

AUC of 0.78, whereas ensemble methods in [22] reached 0.854 for DM prediction, showing how flexible, non linear models can better capture complex interactions among ageing related predictors.

2.5.3 Performance and Validation

Predictive performance across studies is generally moderate to strong, with AUC values between 0.70 and 0.90 for outcomes like DM, UGS, depression, and mortality. This indicates that models can distinguish reasonably well between individuals who will or will not experience an outcome.

However, there are important limitations. Many studies rely only on internal validation, testing models on the same data used for training. This can inflate performance estimates and reduce confidence in applying the model to new populations. To add, most studies are cross sectional, so causal or temporal effects cannot be assessed. When combined with the underuse of nutrition and SOC predictors, these issues indicate that current models may not fully represent the multidimensional nature of ageing and could produce context specific results rather than generalisable predictions [32].

2.5.4 Explainability in ML for ageing

As ML models become more complex, interpretability emerges as a critical challenge. Explainability tools like SHAP can highlight feature importance, but their use is inconsistent, particularly in deep learning models [4, 6, 22]. These methods improve transparency and can highlight unexpected drivers of outcomes, such as psychosocial variables being stronger predictors than biomedical ones, yet black box concerns remain, especially for deep learning models, where feature interactions are difficult to track. For ageing research, where clinical adoption requires both accuracy and trust, explainability is not optional but essential. Current efforts show promise but remain fragmented, with no consensus on best practice. This gap links directly to the broader methodological limits discussed in the next section on gaps in the literature.

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2.6 Gaps in the Literature

First, most studies rely on TILDA or similar cohorts of Irish and predominantly older adults (50+). This limits confidence in how well results transfer to other populations. This thesis focuses on TILDA studying ageing in depth but also makes use of the harmonised dataset, which links TILDA with comparable UK and US cohorts. This helps place findings in a broader international context while retaining a core emphasis on older adults.

Second, there is an over reliance on physical health predictors such as comorbidity counts and mobility measures, while nutrition, SOC, religious or spiritual, and environmental variables are often overlooked. As shown in [11], nutritional biomarkers like lutein and zeaxanthin are strong predictors of FR, yet these are rarely integrated into broader ML pipelines. By contrast, studies such as [27] highlight the role of SOC support and functional decline in malnutrition risk. Together, these examples show that excluding nutrition and SOC factors risks producing models that reinforce the same biomedical biases seen in regression heavy work. From a Bayesian viewpoint, the omission of key features reduces the explanatory power of the model and weakens the posterior probability $P(\text{Outcome} \mid \text{All Features})$ compared to $P(\text{Outcome} \mid \text{Limited Features})$. Including a richer set of features increases the likelihood that the model accurately reflects the true data generating process.

Third, methodological choices remain conservative. Regression models remain dominant, likely due to their interpretability and ease of implementation. However, they often fail to capture non linear interactions and higher order dependencies present in ageing related data. Comparisons with more advanced approaches such as DNNs or transformers are rarely explored. For instance, [41] achieved an AUC of 0.78 using logistic regression for mortality prediction in TILDA, whereas [22] reported an AUC of 0.854 with ensemble methods for DM incidence. This contrast illustrates that more flexible ML models can provide superior predictive performance while maintaining interpretability when combined with explainability techniques. Where else ensembles or brainPAD approaches have been attempted, validation is almost always internal, with little cross cohort testing. Weak validation and limited feature sets reduce the generalisability of the

estimated conditional probabilities and increase the risk of overfitting. Maximum likelihood estimation and Bayesian inference can provide a formal framework for assessing model fit and updating predictions as more diverse and representative data become available.

Fourth, many outcomes relevant to older adults such as pain, sleep quality, and subjective wellbeing remain underexplored. For example, [11] focused on plasma biomarkers for FR without assessing sleep or pain, and UGS models such as [6] similarly omit subjective quality of life. Omitting these outcomes limits the real world application of predictive models. Including broader outcomes in ML pipelines can better capture holistic ageing trajectories and improve interpretability for interventions targeting older adults.

Fifth, cross sectional bias and data handling weaken reliability. Much nutritional work, such as [11], relies on single time point biomarker data, which cannot capture longitudinal change. By contrast, studies applying multi wave TILDA data [26] demonstrate the added value of modelling temporal dynamics. Similarly, missing data are often handled through listwise deletion, as in [41], which risks biasing results toward healthier participants, whereas more robust imputation approaches remain underused. Missing or poorly handled data can distort the likelihood function and hence the posterior predictions, reducing confidence in the model’s outputs.

Finally, interpretability challenges persist. While tools like SHAP have been applied to highlight feature importance [4, 22], their use is inconsistent, and consensus on best practice has yet to emerge. Without transparent explanations, even accurate models may face barriers to clinical acceptance. Taken together, these examples show that ageing research continues to rely on narrow predictors, conservative models, and weak validation. Addressing these gaps requires integrative, diverse, and transparent approaches that link multidimensional predictors with explainable ML.

2.7 Chapter Summary

This chapter reviewed evidence on ageing and chronic disease across three domains, FR, physical function, MH, nutrition, and biological ageing. We also

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examined how machine learning has been applied to predict later life outcomes. The findings showed that while FR is partly reversible, it remains strongly tied to mortality, that MH both shapes and is shaped by physical decline, and that nutrition interacts with biological, metabolic, and SOC systems in ways that are often underestimated.

While machine learning has improved prediction accuracy in ageing research, challenges remain in integrating diverse predictors and ensuring interpretability and longitudinal validation.

Addressing these challenges requires the integrative, longitudinal, and transparent approaches that combine diverse predictors. These gaps motivated the feature engineering and modelling strategies implemented in Chapters 3 and 4, including the use of harmonised variables to ensure reproducibility and generalisability, for ML modelling.

Chapter 3

Technical Approach

3.1 Study Design

TILDA provides longitudinal data across five waves (Waves 1–5) for community dwelling adults in Ireland. For contextual comparison and benchmarking against international cohorts see Chapter 1. An external, harmonised TILDA dataset, was also inspected and prepared. The harmonised dataset, which has been pre aligned to international standards e.g., the Gateway to Global Ageing Data (G2G), contains a subset of variables mapped to a common cross study format and thus differs from the raw TILDA files. Although the harmonised dataset was simpler to process, it was still subjected to the variable filtering and numerical standardisation steps developed as part of this research to ensure compatibility with the five wave cleaned datasets prior to modelling.

3.2 Cohort Definition

All participants who provided data in at least one TILDA wave were considered eligible. To maximise the available sample for machine learning, no exclusions were applied on the basis of age, medical conditions, mobility, cognitive status, or health centre attendance.

Participants were removed only in the rare case that, following manual variable filtering, no analytically usable variables remained for that individual. Filtering

3. TECHNICAL APPROACH

occurred primarily at the *variable level* during the manual screening stage, which involved features removed for extreme missingness, lack of variation, or non informative structure. This approach preserves the largest possible sample while ensuring all retained inputs are analytically meaningful for downstream analysis and model development.

- **Participant inclusion:** All respondents who contributed data in at least one wave were included in the initial sample.
- **Usable data requirement:** Participants were excluded only if all retained variables were missing for that individual after manual variable screening.
- **Longitudinal retention:** Individuals were retained regardless of how many waves they contributed to, enabling initial maximum dataset utilisation.
- **No clinical exclusions:** No participant was removed due to medical diagnoses, cognitive scores, functional limitations, or non attendance at the health centre assessment.

This approach prioritises data integrity and sample size for machine learning applications, differing from exclusion-heavy cohort definitions often used in traditional epidemiological studies [42].

3.3 TILDA data collection and screening

TILDA data collection occurred over five longitudinal waves as detailed in Table 3.1.

Each wave included three primary components:

1. **Computer Assisted Personal Interview (CAPI):** Home based, computer assisted personal interviews covering health, socio demographics, and family.
2. **Self Completion Questionnaire (SCQ):** Paper based self completion questionnaires capturing sensitive information on mental health, lifestyle, and wellbeing.

3.3 TILDA data collection and screening

3. **Health Assessment:** Clinical and biomarker measurements conducted in dedicated centres or at home (Wave 1 and Wave 3).

All TILDA data were pseudonymised prior to release, meaning that personal identifiers such as names and addresses were removed or replaced with unique codes to protect participant confidentiality. Each participant received a unique identifier ('id'), along with household and cluster identifiers ('household', 'cluster'), allowing longitudinal linkage across waves without revealing individual identities.

Pseudonymisation ensures that analyses comply with data protection standards while retaining the structural and analytical integrity of the dataset. From a data cleaning and machine learning perspective, pseudonymised identifiers enable accurate merging of data across waves, preserve the longitudinal design, and support reproducible preprocessing and feature engineering. While these identifiers cannot be used for analyses of personal demographics like names or addresses, this limitation does not affect the modelling of health outcomes, lifestyle factors, or other variables of interest.

Prior to any data cleaning or modelling, a structured multi stage workflow was implemented to screen and reduce the dimensionality of the raw TILDA datasets. This workflow separates automated diagnostic profiling from manual variable exclusion and ensures transparency in all variable selection decisions.

The process consisted of the following steps:

1. **Raw data loading:** Raw TILDA PMF files were loaded without cleaning or recoding. Waves 1–5 were loaded from Statistical Package for the Social Sciences (SPSS) (.sav) files, while the harmonised dataset was loaded from a (Stata) (.dta) file due to technical compatibility issues.
2. **Automated variable profiling (Python):** For every variable, summary statistics were computed programmatically, including data type, proportion of missing values, number of unique values, most frequent value, and frequency of the most frequent value. These summaries were exported to spreadsheet format for inspection.

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3. **Manual variable screening (spreadsheet based):** Using predefined rule based thresholds, variables were manually reviewed and flagged for exclusion based on missingness, lack of variation, dominance of a single category, or administrative function.
4. **Creation of retained variable lists:** Variables not flagged for exclusion were programmatically concatenated in `.xlsx` and manually transferred into wave specific JavaScript Object Notation (JSON) files representing the retained variables of interest.
5. **Reloading filtered datasets:** The JSON files were then used to subset the original raw datasets, producing reduced datasets with identical rows but fewer variables. No cleaning, recoding, or imputation was applied at this stage.

These reduced datasets form the direct input to the automated preprocessing and modelling pipeline described in Chapter 4.

Table 3.1: Dimensions of raw TILDA and harmonised datasets (prior to filtering).

Dataset	Collection Period	Participants (N)	Raw Variables (N)
Wave 1	2009–2011	8501	1619
Wave 2	2012–2013	7206	724
Wave 3	2014–2015	6397	1029
Wave 4	2016	5713	990
Wave 5	2018–2019	4978	983
Harmonised	2016 (release)	8501	572

Following the five step workflow outlined above, Python scripts were used to profile all variables across each wave. For every variable, the following metrics were computed: data type, number of missing values, number of unique values, most frequent value, and frequency of the most frequent value. These summaries were exported to spreadsheet format to support structured manual review prior to variable dropping.

Manual filtering was applied according to the following criteria:

3.3 TILDA data collection and screening

- **Missingness:** Drop variables with $> 80\%$ missing values. This threshold balances retaining sufficient feature coverage with ensuring statistical reliability. Variables exceeding this threshold provide limited information in most cases, increase imputation uncertainty, and reduce effective sample size, which is particularly critical in longitudinal machine learning analyses.
- **Unique values:** Drop variables with a single observed value, as they contain no discriminative information and cannot contribute to modelling or inference. Variables with low cardinality (2–4 unique values) were retained for review, as they frequently represent meaningful categorical responses (e.g., binary indicators or ordinal scales) relevant to ageing outcomes.
- **Dominant category (top frequency):** Variables where a single category (sample value) occurs in over $> 90\%$ of observations were flagged. Such dominance indicates limited variability, reducing analytical utility and predictive value. Variables exceeding this threshold were marked for review or exclusion.
- **Identifiers / administrative fields:** IDs, version tags, and sampling weights were identified using the PMF metadata. Variable names and descriptions were exported from each wave’s PMF file into CSV files (code provided in Appendix A) to systematically document and review features prior to filtering. These exports enabled structured manual inspection and traceability between raw PMF metadata and downstream feature selection decisions.

As part of the manual screening and filtering workflow. Variables exceeding exclusion thresholds (e.g., high missingness, single value, dominant category, administrative identifiers) were visually flagged (e.g., marked in red). Non flagged variables were programmatically concatenated using spreadsheet formulas (e.g., `=TEXTJOIN(", ", TRUE, J2:Jn)`) and then manually transferred into wave specific JSON files for each wave. These files represent the variables of interest retained after manual screening and filtering and serve as inputs for the upcoming automated data preprocessing & cleaning in Chapter 4.

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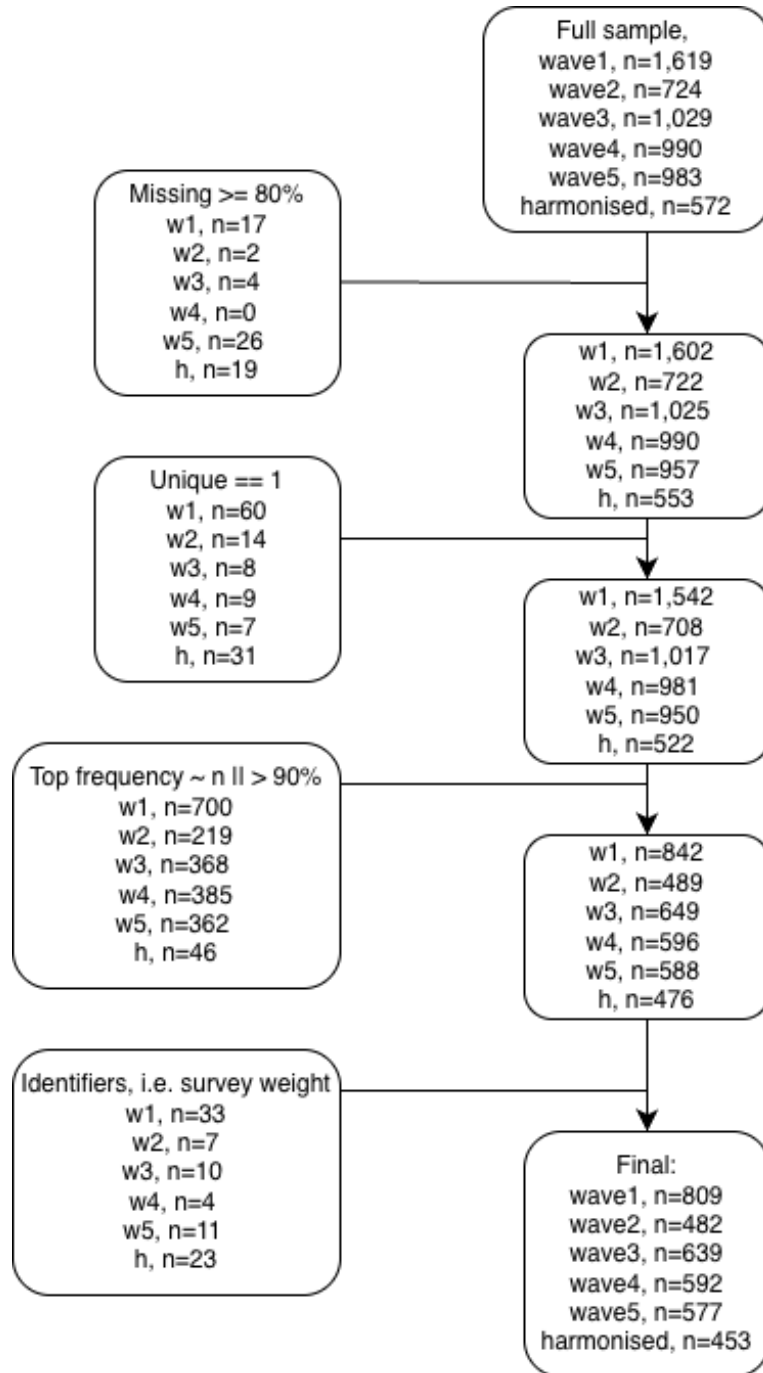


Figure 3.1: Flowchart of the five step workflow for variable screening and filtering. Placeholder counts indicate the progression of remaining variables per dataset from raw to filtered datasets after each step, ready for automated preprocessing.

The retained variables, defined in JSON files, were subsequently used to subset the raw datasets. This produced reduced CSV datasets per wave, containing only the variables of interest. At this stage, no cleaning or recoding was performed. The operation served purely as dimensionality reduction, preparing the data for the automated nine-scenario cleaning pipeline described in Chapter 4.

Importantly, all variable exclusion decisions were completed at this stage. Subsequent preprocessing steps focus exclusively on numerical standardisation, recoding, and preparation for machine learning rather than further feature removal. With the variables of interest selected and the datasets reduced to a manageable dimensionality, the next stage focuses on automated preprocessing, recoding, and preparation for machine learning.

3.4 Thematic Grouping

After manual filtering, the retained variables were organised into seven thematic domains based on ageing literature and study objectives. Each variable's prefix, as defined in the PMF metadata (Table 3.2), indicates its source or research domain. This systematic assignment ensures that each thematic group contains conceptually related features and provides a clear structure for downstream automated preprocessing and machine learning.

Each variable in TILDA was assigned a prefix indicating its data source or research domain. Home based CAPI variables use a two letter module code followed by a three digit question code (e.g., `ph121`). Variables with multiple responses or loops include an underscore and number to specify the response or iteration (e.g., `ph201_05`).

Variables from the self completion questionnaire are prefixed with SCQ, while derived variables constructed from either CAPI or SCQ data use uppercase codes corresponding to their research area (Table 3.2). These prefixes provide a consistent framework for categorising features and guiding the assignment into thematic domains for downstream analysis.

Using the PMF prefixes as a guide, variables were assigned to seven thematic groups (Table 3.3), each capturing a conceptually coherent domain: physical,

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Table 3.2: PMF Derived variable prefixes grouped by thematic domains [3].

Variable Prefix	Research Area
1: Socio Demographic & Context	
INC	Income variables
SES	Social class
SOC	Social
2: Physical Health & Clinical Baseline	
BP	Blood pressure from sphygmomanometer
MD	Medication data
SCR	Screening tests
3: Functional Status & Frailty	
ADL	Activities of daily living
DIS	Disability, functional impairment, helpers
FR	Frailty, gait, exercise, falls and fractures
GRIP	Grip strength
IADL	Instrumental activities of daily living
4: Lifestyle & Behavioural	
BEH	Health behaviours
5: Nutrition & Metabolic Biomarkers	
BLOODS	Results of blood sample analysis
6: Psychological & Cognitive	
COG	Scales and questions on cognition
CRT	Choice reaction time (cognitive test)
MH	Scales and questions on mental health and wellbeing
Other / Administrative	
G2G	Gateway to Global Aging Data

cognitive, social, and behavioural aspects of ageing. This structure supports reproducible preprocessing and helps machine learning models focus on meaningful, longitudinally robust features.

Specifically, the thematic grouping:

1. Provides a clear framework for automated cleaning and filtering rules.
2. Ensures that model input variables are conceptually aligned with study objectives.

The seven thematic groups integrate information from CAPI, SCQ, and derived variables, covering socio demographic context, physical health, functional status, lifestyle behaviours, nutrition/metabolic biomarkers, and psychological/cognitive measures. This structured grouping enhances interpretability, ensures comprehensive domain coverage, and provides a foundation for downstream predictive modelling and risk assessment.

The gradient *heatmap* integrated at the bottom of Table 3.3 visually illustrates the relative dominance of each thematic domain across the five waves. Darker blue shading indicates domains with a higher number of variables, providing a quick summary of domain coverage and prominence. This integrated visualisation supports interpretation by highlighting which domains contribute the most features to the dataset.

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Table 3.3: Thematic domain grouping of longitudinal features.

ID	Thematic Group	Description	Representative Variable Examples	Total N Features
1	Socio-Demographic & Context	Baseline demographic and socioeconomic indicators	Age, sex, education, income, marital status	844
2	Physical Health & Clinical	Clinical and self reported health measures	Blood pressure, BMI, chronic disease status, self rated health	839
3	Functional Status & Frailty	Physical functioning and frailty markers	Activities of Daily Living (ADL)/Instrumental Activities of Daily Living (IADL), grip strength, gait speed	174
4	Lifestyle & Behavioural	Health behaviours and activity patterns	Physical activity, smoking, alcohol use	120
5	Nutrition & Metabolic	Metabolic and biomarker indicators	C-Reactive Protein (CRP), creatinine, lipid markers	22
6	Psychological & Cognitive	Cognitive performance and mental health	Mini-Mental State Examination (MMSE), depressive symptoms, stress scale	291
7	Other / Administrative	Survey metadata and structural variables	sampling weight, completion status, contact names	809

	SocioDemographic_Context	PhysicalHealth_Clinical	FunctionalStatus_Frailty	Lifestyle_Behavioural	Nutrition_Metabolic	Psychological_Cognitive	Other_Administrative	Total_Features
wave1	272	149	48	58	10	97	175	809
wave2	119	148	18	21	0	38	138	482
wave3	164	184	44	14	8	93	132	639
wave4	161	186	30	13	4	36	162	592
wave5	128	172	34	14	0	27	202	577
Total_per_Domain	844	839	174	120	22	291	809	3099

Thematic domain dominance across waves (higher n of features per domain).

Chapter 4

Implementation Details

Following manual variable filtering in (Chapter 3), the reduced datasets contained only the variables of interest. To further understand the characteristics of these retained variables, the datasets were profiled using an exploratory visualiser (`.ipynb`) file implemented with `pandas`, `numpy`, `matplotlib`, and `seaborn`. This step examined data types, missingness patterns, unique value distributions, and representative sample values across all waves. The purpose of this profiling stage was not transformation but to expose structural inconsistencies and latent formatting issues that would undermine direct numerical conversion.

Table 4.1: Input shape for cleaning pipeline (post manual variable filtering).

Dataset	Row (participants)	Column (features)
Wave 1	8501	809
Wave 2	7206	482
Wave 3	6397	639
Wave 4	5713	592
Wave 5	4978	577
Harmonised	8501	453

For example, in Wave 1, the visualiser detected 650 categorical columns, 150 (`float64`) columns, and 9 object columns, while the harmonised dataset contained 304 categorical columns, 148 `float32` columns, and several integer columns. Age distributions confirmed that the study population was restricted

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to older adults across waves, however, ages were bottom coded for participants younger than 50 years and top coded for those aged 65–85 years and above. Missing value summaries indicated substantial variation (e.g., mean missing values ranging from 4.7% in Wave 4 to 35% in the harmonised dataset, with some variables exceeding 77% missing).

Despite this, the exploratory visualiser was limited because the raw data were not yet cleaned, many outputs were incorrect or nonsensical due to hybrid formats and inconsistent coding. For instance, inspection revealed that many variables across both the wave specific and harmonised datasets, while appearing numeric at first glance, in fact contained hybrid representations combining numeric prefixes with descriptive text (e.g., “1.Rarely or none of the time (less than 1 day)”, “4.All of the time (5–7 days)”, or prefixed labels such as “0.not coupled”, “1.coupled”). A trivial string stripping approach would therefore produce incorrect numeric values (for example, reducing “5–7 days” to 5) resulting in silently introducing semantic distortion. Similarly, numeric columns sometimes contained sentinel codes (e.g., -98, -99) or embedded symbols (‘\$’, ‘*cm*’, ‘%’) that skewed visual summaries.

These insights, revealed through systematic profiling, guided the development of the subsequent cleaning pipeline, ensuring consistent numeric logic representations (e.g., midpoints for ranges, standardised sentinel handling).

4.1 Cleaning Engine

The core challenge in converting heterogeneous variable types across the six datasets (Table 4.1) was achieving a single numerical standard suitable for downstream analysis. This required enforcing *numerical homogeneity*, defined here as representing all retained features using a single numerical representation (`float64`) with a unified missing or non response sentinel (`-1.0`). The exploratory profiling across all waves showed that categorical variables were dominant overall (2,916 columns), while numeric variables were fragmented across multiple types, with `float64` being the most prevalent native numeric representation (462 columns), motivating its use as a common standard to ensure consistency and compatibility with downstream ML models, while accommodating decimal values arising

from range midpoints, frequency normalisation, and ordinal scaling that cannot be reliably represented using integer types. To address this, the solution was implemented as a two stage system. The first stage performs automated *variable scenario detection*, identifying the structural form of each variable based on observed value patterns using the `detect_column_scenario_mapping` function (Appendix C). This stage includes additional diagnostic categories used during development to surface unhandled edge cases.

The second stage applies a deterministic *six step numerical cleaning engine* (Appendix E). While nine scenario categories were initially defined during development, three served exclusively as validation fallbacks and were progressively eliminated as detection coverage improved, with a full scenario to step mapping counts provided in Appendix B. The final pipeline therefore consists of six active transformation steps, applied sequentially and uniformly across all waves.

4.1.1 Stage 1: Variable scenario mapping and validation

Stage 1 focused on automatically identifying how each variable should be handled during cleaning. The automated detection function in Appendix C was developed to classify variables based on their observed value structure, including the presence of sentinel codes, numeric annotations, ordinal labels, and mixed string numeric formats. While multiple intermediate detection categories were used during development, these served solely as internal routing logic. While all variables passed sequentially through the six step cleaning pipeline, most were fully resolved in the earlier steps (Steps 1–5). The final step (Step 6 in Appendix D) was applied only to residual variables that could not be deterministically resolved by earlier transformations.

The additional detection categories (e.g., “single value categorical”, “special symbol mixed”, and “unknown/complex”) were intentionally included as diagnostic fallbacks to surface unhandled edge cases due to the complexity and size of the data. As rules matured, variables initially defaulted to these fallback categories were progressively resolved into earlier, well defined scenarios, leaving none remaining for the final pipeline execution.

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To ensure robustness, the classification output was immediately processed by a validation routine that generated a comprehensive dictionary of feature states before and after cleaning, including all representative sample values for each variable. This allowed direct inspection of how the same variable was represented across waves prior to cleaning, particularly in cases where identical semantic categories appeared in different orders or formats. By comparing these grouped samples before and after each cleaning step, classification errors and inconsistent mappings were identified and corrected prior to execution of subsequent transformations.

The dictionary generated inside the `show_grouped_vars_for_scenarios` function (Appendix F) was formatted to create an automatic friendly “.md” file for rapid, surface level verification, which honed the final mapping dictionary (an example can be found in Appendix G) for all features across all waves. This process of using automated detection followed by visual, manual validation was essential for correcting errors in classification logic before the subsequent scenario transformations were executed in the pipeline.

4.1.2 Stage 2: Deterministic six step cleaning sequence

The nine input scenarios were processed sequentially by the *six step cleaning engine*, resolving simple and unambiguous cases first to maximise the proportion of clean numerical features before tackling more complex structures, particularly censored and multi-class scenarios (Figure 4.1). With later steps progressively shifting from cleaning into semantic feature engineering, this deterministic sequence applied identical transformation and feature encoding logic to each wave, with later steps (particularly Steps 4–6) performing rule-based feature engineering by deriving semantically meaningful numerical representations directly from observed category tokens at runtime. The methodological rationale and implications of this integrated engineering which reframes cleaning as semantic feature construction are detailed in Section 4.1.3. (See Appendix D for implementation).

This design reflects the empirical diversity of survey encodings, where a small subset of variables only reveal their semantic structure after simpler transformations have been applied.

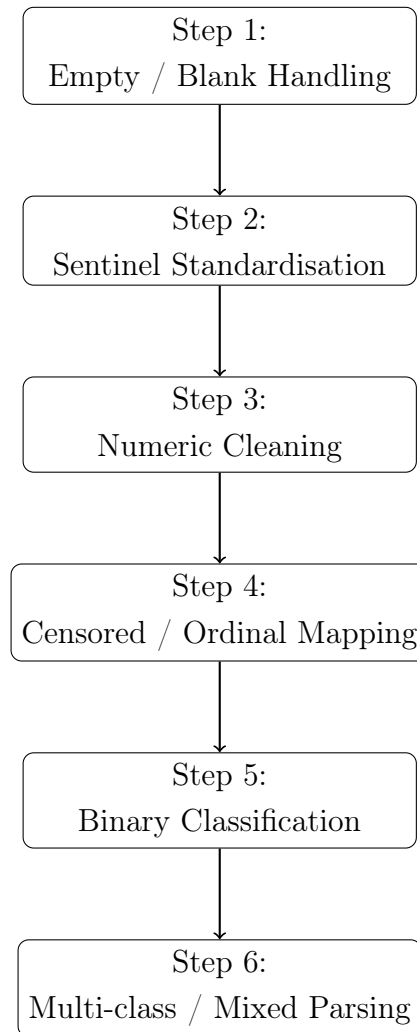


Figure 4.1: Flowchart of the final six step cleaning pipeline applied to all TILDA waves. See full transformation tables in Appendix B.7 and E for a detailed progression “.ipynb” script at each scenario cleaning step.

4. IMPLEMENTATION DETAILS

The final cleaning pipeline consisted of six deterministic steps applied uniformly across all TILDA waves, with increasingly complex transformations handled in later stages. Steps 4–6 required particular attention due to hybrid, ordinal, or multi-class formats. A summary of each step and examples is provided below.

1. **Empty / Blank Values:** All missing entries, empty strings, and white-space were converted to the standard non-response sentinel (-1.0).
2. **Sentinel Values:** Wave specific non-response codes (e.g., -98, -99) and textual sentinels (e.g., "dk", "not applicable") were mapped to the unified (-1.0) placeholder.
3. **Numeric Transformation:** Numeric like columns containing embedded units, symbols, or hybrid strings (e.g., 50+, <10, 1-2.5) were parsed using regex and converted to `float64`. For ranges, midpoints &/or boundary values were used. For censored or annotated values, appropriate upper/lower bounds were applied.
4. **Censored / Ordinal Variables:** This step addressed variables representing ordinal scales, bracketed numeric ranges, or frequency expressions (e.g., "2–3 days a week", "daily"). Transformations included:
 - Converting textual ordinal labels and Likert scale responses to numeric equivalents.
 - Mapping calendar based frequency descriptors to numeric weekly equivalents.
 - Manual verification for low count or unusual variables, supported by the Markdown audit outputs (Appendix G), to ensure semantic meaning was preserved.
5. **Binary Categorical Variables:** Two-level features (e.g., Yes/No, Male/Female) were dynamically encoded using mappings derived from the observed value set of each variable:
 - Semantic rules were applied e.g., ("no", "not", "female" → 0.0) and ("yes", "male" → 1.0).

- Count based detection verified that variables had exactly two non-missing unique values.
- Features with more than two sample values were also analysed for sentinel presence:
 - If a variable had three observed values and one was a sentinel, it was treated as binary.
 - If a variable had four observed values and two were sentinels, it was treated as binary.
- Sentinels were preserved as -1.0.
- This approach ensured order invariant encoding across waves, even when original labels or numeric prefixes differed.
- *Example (sentinel case)*: Feature `f1002_5` with pre transformed values ['NOT Getting in or out of bed', -1.0, 'Getting in or out of bed'] was encoded as [0.0, -1.0, 1.0], demonstrating how a sentinel preserves binary semantics.

6. Multi-class / Mixed Variables (rule based feature engineering):

Variables with remaining multi-class structure, mixed types, or unresolved string numeric hybrids after earlier transformations were processed in the final step.

Technical implementation details for this step are provided in Appendix D:

- Lexical parsing extracted numbers, ranges, units, percentages, and textual frequency descriptors into floats, yielding interpretable numerical feature representations.
- Sentinel handling ensured all known non-response codes mapped to (-1.0).
- Original values were grouped and assigned numeric identifiers in first seen order, non numeric categorical labels were assigned integer codes above the highest observed numeric value to prevent collisions.
- Manual inspection using the Markdown audit file verified that semantic equivalence was maintained across waves.

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- Step 6 was designed to be *order invariant*, dynamically interpreting category values within each wave rather than relying on predefined lookup tables shared across waves. This preserved consistent numeric representation regardless of variable ordering or wave specific encodings.
 - *Example:* Step 5 handled simple binary variables such as `CHRany_pain` (any pain reported) with pre transformed values [`'pain'`, `'No pain'`] and numeric mapping [`1.0`, `0.0`]. In contrast, Step 6 dynamically processed multi-class variables like `we114` (pre tax wage/salary) many of which contained over 20 distinct pre transformed categories (e.g., `-1.0`, `'200-299'`, `'Less than 100'`, ..., `'6,000-6,999'`) into consistent numeric representations (e.g., `-1.0`, `249.5`, `99`, ..., `6499.5`). This illustrates why Step 6 required wave specific, dynamic parsing to maintain semantic equivalence across diverse observed values.

4.1.3 Feature engineering within the cleaning pipeline

Although the six step pipeline is framed primarily as a numerical cleaning procedure, Steps 4–6 also perform explicit rule-based feature engineering by interpreting heterogeneous categorical survey responses and converting them into semantically meaningful numerical representations. Rather than acting as a separate post processing stage, feature engineering is embedded directly within the cleaning logic, formalising ordinal structure, frequency, and magnitude encoded in text into directly modelable `float64` values suitable for downstream analysis.

Crucially, this process engineers the *representation* of each feature rather than expanding the feature space. For example, free text frequency responses such as “once or twice a week”, “a couple of times per month”, or “almost every day” are not treated as nominal categories. Instead, lexical cues (e.g., *once*, *twice*, *couple*) are resolved to numeric quantities and normalised onto a common temporal scale, yielding a continuous rate representation. Implicit semantic information such as ordering, intensity, and frequency is therefore preserved through deterministic parsing and normalisation without introducing additional variables. This

design maintains the original dimensionality of the dataset while ensuring numerical homogeneity and semantic integrity, producing a compact and model ready representation of complex real world survey data.

4.1.4 Cross wave semantic verification

Given the absence of guaranteed category ordering across waves, preserving semantic equivalence during numerical encoding and feature representation was essential to avoid introducing distortions into downstream ML modelling. Pre and post transformation values for each group were exported to a readable Mark-down (.md) file, as shown in Appendix G. This representation confirmed that features of known importance for downstream predictive models were preserved and correctly encoded. For example, features such as **ph001** (self-rated general health), **ph009** (comparative self-rated health), and **bh006** (smoking frequency) correspond to items identified by [16] as among the most influential predictors in mortality models by the FI. Retaining these features from the manual filtering thresholds applied in (Chapter 3) and encoding them consistently across waves ensured that the cleaned dataset maintains coverage and semantic integrity, preventing distortions that could potentially compromise model performance.

A key challenge in cleaning the five TILDA waves was the lack of guaranteed ordering for categorical values of the same variable across different waves. Table 4.2 illustrates this for Wave 4 and Wave 5.

For example, gender (**sex**) variables appeared under different value encodings, with varied category orderings and numeric prefixes across waves. Values such as ['Male', 'Female'], ['Female', 'Male'], and prefixed forms like ['1.male', '2.female'] were all mapped to a consistent numeric representation (1.0 for male, 0.0 for female). This ensured that semantic equivalence, rather than positional ordering or wave specific encoding, determined numerical assignment.

The variable **ph001** found in Appendix G corresponds to the survey question: "Now I would like to ask you some questions about your health. Would you say your health is...?". Responses are ordinal (Poor to Excellent) with non-response categories.

4. IMPLEMENTATION DETAILS

The grouped Markdown representation reveals three critical points:

1. Semantically identical categories are correctly grouped even under different variable names (ph001 vs ph009), enabling direct comparison of representations that would otherwise appear fragmented across waves.
2. Numerical encoding is invariant to category ordering, preserving ordinal meaning across waves. Without this approach, positional encoding would assign inconsistent values, potentially distorting performance for downstream models.
3. For variables with a large and heterogeneous set of observed values (e.g., income bands, frequency ranges, or mixed numeric text responses), the grouped view provided essential visibility into how complex category spaces were being collapsed into stable numeric representations, supporting both validation and error detection at scale.

Table 4.2: Inconsistent category ordering for the same feature *ph009* (self-rated health compared to other people your age) across waves and its impact on positional (order-based) label encoding.

Pos	Wave 4		Wave 5	
	Category	Code	Category	Code
1	Good	3.0	Very Good	4.0
2	Fair	2.0	Fair	2.0
3	Poor	1.0	Excellent	5.0
4	Excellent	5.0	Good	3.0
5	DK	-1.0	Poor	1.0
6	Very Good	4.0	DK	-1.0
7	RF	-1.0	RF	-1.0

4.2 Validation of the cleaning pipeline

The purpose of the cleaning validation stage was to assess whether the automated numerical parsing and encoding pipeline preserved meaningful biological and statistical structure within the cleaned datasets. Rather than asserting correctness based solely on successful execution, validation was performed by evaluating whether known relationships reported in the literature could be recovered using the cleaned data under realistic data governance constraints.

Due to restricted access to certain outcome variables within TILDA, full end-to-end replication of all published models was not always feasible. In particular, access through the **Hotdesk Facility** for the official mortality outcomes is subject to additional governance requirements, including institutional GDPR training certification and approval by the TILDA data access committee [43].

As a result, validation focused on two complementary strategies:

- (i) Internal consistency checks against established biological and clinical relationships, and
- (ii) Constrained replication of published modelling frameworks using available predictors and the derived `8y_mortality_proxy` outcome.

Together, these validation steps demonstrate that the cleaning pipeline retains expected relational patterns and produces datasets appropriate for downstream modelling tasks.

4.2.1 Internal feature relationships

The first validation stage assessed whether cleaned predictors preserved biologically and clinically plausible relationships reported in prior TILDA studies. This analysis focused on predictors identified by [4] as accounting for approximately 95% of the explained variance in GSR, defined as the difference between maximum and usual gait speed.

As GSR is not directly available in the public Wave 3 dataset, validation was performed by examining the directional relationships between the 7 primary predictors themselves. This approach evaluates whether the numerical encoding

4. IMPLEMENTATION DETAILS

process, including sentinel handling, missing value treatment, and ordinal mapping, retained the expected internal structure of the data rather than reproducing outcome specific performance.

Figure 4.2 presents the correlation matrix for the selected predictors following cleaning. Grip strength exhibited a negative association with both age and chair stand time, reflecting the inverse relationship between muscular strength and physical performance decline. Cognitive performance (Montreal Cognitive Assessment (MOCA)) demonstrated a moderate negative correlation with age and a positive association with educational level, consistent with known patterns of cognitive reserve in older adults. Education correlated positively with cognitive performance, and BMI showed the expected weak negative association with age, reflecting reduced physiological reserve in older cohorts.

Table 4.3 summarises the expected and observed directional relationships between key predictor pairs. Notably, grip strength showed moderate negative correlations with both age ($r = -0.265$) and chair stand time ($r = -0.170$), reflecting the expected inverse relationship between muscular strength and physical performance decline. Cognitive performance (MOCA) displayed a moderate negative association with age ($r = -0.381$) and a positive association with education ($r = 0.367$), consistent with known patterns of cognitive reserve in older adults. BMI demonstrated the anticipated weak negative correlation with age ($r = -0.073$). These results confirm that the cleaning pipeline retained the expected numerical structure and relational patterns of the predictors, rather than introducing artificial associations.

4.2.2 Model based validation

To evaluate whether the cleaned dataset preserves predictive utility, a constrained replication of [16] was conducted. The resulting predictive performance for both model variants and sexes is summarised in Table 4.4. The study predicted 8-year mortality using a 32-item FI. In the cleaned Wave 1 dataset, only 16 of these 32 FI items remained following the manual variable filtering described in Chapter 3, which prioritised global numeric consistency across waves.

4.2 Validation of the cleaning pipeline

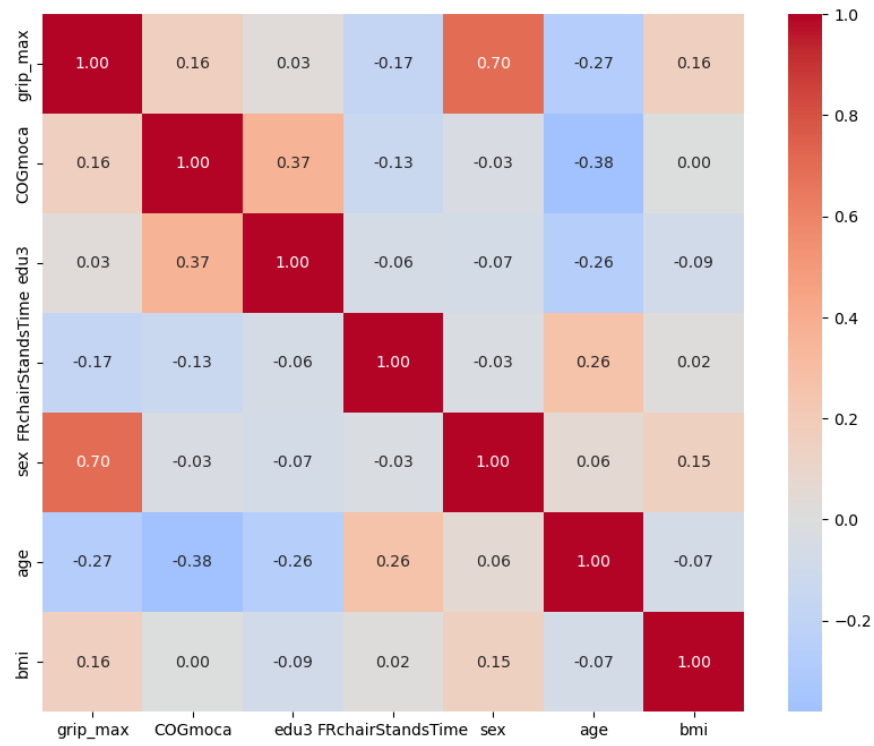


Figure 4.2: Correlation heatmap for top 7 predictors (Wave 3).

4. IMPLEMENTATION DETAILS

Table 4.3: Directional validation of the 7 features explaining 95% of GSR variance as identified by [4].

Predictor 1	Predictor 2	Expected Direction / Justification	Observed r
Grip Strength (GRIP)	Age	Negative (-) ; GRIP is a positive predictor of GSR, while age is a negative predictor [4].	-0.265
GRIP	Chair Stands Time	Negative (-) ; Higher strength (grip) is expected to correlate with faster (lower) performance time in physical tests [4].	-0.170
MOCA Score	Age	Negative (-) ; Cognitive performance (MOCA) contributes positively to GSR, whereas increasing age is associated with reduced reserve [4].	-0.381
Education	MOCA Score	Positive (+) ; Third level education and cognitive scores are both cited as positive contributors to cognitive reserve and GSR [4].	0.367
BMI	Age	Weak / Mixed ; BMI and age both negatively impact GSR, though their internal correlation in older cohorts is typically weak [4].	-0.073

4.2 Validation of the cleaning pipeline

A ‘8y_mortality_proxy’ outcome was constructed to approximate the sex specific 8-year mortality rates reported in [16] (men: 14.9%, women: 11.1%) as observed via linkage to Ireland’s Central Register Office records [44]. Due to access restrictions, the official mortality data were unavailable; therefore, the proxy was created by randomly selecting deceased participants from the Wave 1 cohort aged 50 or older to match these percentages observed at Wave 5 follow up, and randomly sampling an equal number of alive participants to generate a balanced dataset for model training. The resulting datasets included 1,114 men (557 deceased) and 982 women (491 deceased), reflecting the baseline distribution of the Wave 1 participants (≥ 50 years) = 8,172: 3,743 men and 4,429 women. This proxy serves a diagnostic role: although it does not replicate true mortality events, it allows evaluation of whether numerical relationships and predictive information are preserved in the cleaned data.

Two model variants were tested for each sex:

1. **FI only model:** using the 16 available FI items, without age.
2. **FI + Age model:** including age as an additional predictor.

Model training and evaluation followed the procedure reported in [16] as closely as permitted by the available data. Features were min–max normalised prior to modelling. For each sex, data were split into training (80%) and testing (20%) sets. To mitigate overfitting and obtain robust estimates of model performance, 5-fold stratified cross-validation was applied on the training data. AUC was selected as the primary evaluation metric due to its widespread use in prior TILDA and ML based studies (Figure 2.2) and its robustness to class imbalance.

For six FI items (mh0014, ph001, ph102, ph108, ph114, bh201), the cleaned dataset transformations differed from the scoring rules published in [16]. To align precisely with the original 16/32-item FI methodology, these items were recoded following the published ordinal and dichotomous mappings. This step was necessary to ensure that the partially remapped FI reflects the same interpretation of health deficits used in the literature and is consistent with standard FI construction principles [45]. Models were then re-evaluated using this partial remapping FI to assess sensitivity to feature encoding.

These results highlight several key points:

4. IMPLEMENTATION DETAILS

Table 4.4: Model replication results: comparison of predictive performance using the cleaned and 6 recoded FI features with the constructed `8y_mortality_proxy`.

Model	Cleaned FI (16 items)		Recoded FI (paper[16] aligned)	
	Men AUC	Women AUC	Men AUC	Women AUC
FI-only	0.5068	0.5118	0.5672	0.5368
FI + Age	0.5156	0.5312	0.5723	0.6052

1. Despite relying on a reduced subset of the original 32 FI items (16 available in the cleaned dataset), the `8y_mortality_proxy` models reflect expected trends. In particular, the AUCs for the FI only model (men: 0.507, women: 0.512) are lower than those reported by [16] (men: 0.70, women: 0.70), which is consistent with the use of a proxy mortality outcome for participants. Including age improves predictive performance (men: 0.516, women: 0.531), reproducing the pattern observed in the original study (men: 0.80, women: 0.90). The sex specific distributions FI in our dataset (men: 0.2254–0.2258, women: 0.243–0.258 across alive/deceased) are broadly aligned with the published values (men: 0.11, women: 0.13).
2. Re-mapping the six items to match the [16] FI scoring yielded modest increases in AUC scores (men: 0.567, women: 0.537 for FI only; men: 0.572, women: 0.605 for FI + age), illustrating that partial recoding restores some of the predictive information lost during initial variable filtering. This highlights the sensitivity of predictive performance to encoding choices and demonstrates that oversimplified 0 – 1 scaling of metrics with scientifically constrained ranges, such as FI where 0 indicates minimal deficits and 1 indicates maximal deficits, can distort interpretation and compromise the meaningfulness of derived scores.
3. The cleaning pipeline successfully produced a fully numeric dataset suitable for downstream machine learning while maintaining sample characteristics close to the original cohort. Total participants over the age of 50 with a sex distribution of 45.8% men and 54.2% women, and expected deaths based on published rates of 14.9% (men) and 11.1% (women), confirming that

4.2 Validation of the cleaning pipeline

the preprocessing and filtering preserved key demographic and outcome structures.

Chapter 5

Empirical Evaluation

You can also use this kind of referencing if needed in the text:

Chapter 6

Conclusion

6.1 Summary

6.2 Conclusions

6.2.1 Data availability Statement

The data analysed in this study is subject to the following licenses/ restrictions:
The datasets generated during and/or analysed during the current study are not publicly available due to data protection regulations but are accessible at TILDA on reasonable request. Requests to access these datasets should be directed to <https://tilda.tcd.ie/data/accessing-data/>.

6.2.2 Ethics Statement

The studies involving human participants were reviewed and approved by Faculty of Health Sciences Research Ethics Committee at Trinity College Dublin, Ireland. The patients/ participants provided their written informed consent to participate in this study.

6. CONCLUSION

6.3 Contributions

6.4 Future Work

Appendix A

Python code for exporting PMF metadata to CSV

```
1 import pandas as pd
2 import pyreadstat
3 from pathlib import Path
4 import json
5
6 data_files = {
7     "wave1.json": "data/0053-01_TILDA_Wave1_2009-2011/0053-01
8         _TILDA_Wave1_Data/SPSS/0053-01_TILDA_Wave1_PMF_v1.12.sav",
9     "wave2.json": "data/0053-01_TILDA_Wave2_2012-2013/Data
10         /0053-02_TILDA_Wave2_PMF_v2.7.sav",
11     "wave3.json": "data/0053-01_TILDA_Wave3_2014-2015/Data
12         /0053-04_TILDA_Wave3_PMF_v3.6.sav",
13     "wave4.json": "data/0053-01_TILDA_Wave4_2016/Data/0053-05
14         _TILDA_Wave4_PMF_v4.4.sav",
15     "wave5.json": "data/0053-01_TILDA_Wave5_2018/Data/0053-06
16         _TILDA_Wave5_PMF_v5.5.sav",
17     "harmonized.json": "data/0053-06_TILDA_Harmonized_2016/Data
18         /0053-03_TILDA_Harmonized_v1.2.dta",
19 }
20
21 def load_df_with_var_of_interest(
22     var_dir="var_of_interest", data_files=data_files, verbose=
23         False
```

A. PYTHON CODE FOR EXPORTING PMF METADATA TO CSV

```
18 ):
19     dfs = {}
20     var_dir = Path(var_dir)
21
22     for json_name, data_path in data_files.items():
23         file = var_dir / json_name
24         with open(file, "r") as f:
25             vars_list = json.load(f).get("variables_of_interest")
26             if verbose:
27                 print(f"Variables_of_interest_from_{file.name}:_{vars_list[:10]}_...")
28
29         if data_path.endswith(".sav"):
30             df_full, meta = pyreadstat.read_sav(
31                 data_path, encoding="latin1", apply_value_formats=True
32             )
33             df, _ = pyreadstat.read_sav(
34                 data_path,
35                 usecols=vars_list,
36                 encoding="latin1",
37                 apply_value_formats=True,
38             )
39         else:
40             df_full = pd.read_stata(data_path,
41                                     convert_categoricals=True)
42             df = pd.read_stata(data_path, convert_categoricals=True)[vars_list]
43
44         if verbose:
45             print(
46                 f"Data_shape_(Original):_{df_full.shape[0]}_rows_x_{df_full.shape[1]}_columns"
47             )
48             print(f"Data_shape_(Cleaned):_{df.shape[0]}_rows_x_{df.shape[1]}_columns")
49             print("-----")
50
51     dfs[file.stem] = df
```

```

52     return dfs
53
54
55 if __name__ == "__main__":
56     for json_name, path_str in data_files.items():
57         path = Path(path_str)
58         wave_name = json_name.replace(".json", "")
59         if path_str.endswith(".sav"):
60             # SPSS
61             df, meta = pyreadstat.read_sav(
62                 path, encoding="latin1", apply_value_formats=True
63             )
64
65         else: # Stata .dta file
66             df, meta = pyreadstat.read_dta(path, encoding="latin1")
67
68         pd.DataFrame(
69             {"variable": meta.column_names, "label": meta.
70              column_labels}
71         ).to_csv(Path("output") / f"{wave_name}
72                  _variable_descriptions.csv", index=False)
73
74         print(f"-----wave_{path}-----")
75         print(df.shape)
76         print(df.columns[:20])
77         print(
78             "Variable/Feature_descriptions, csv_file_saved_
79             successfully->dir=output/"
80         )
81         df.to_csv((Path("output") / f"{wave_name}.csv"), index=
82                   False)
83
84     dfs = load_df_with_var_of_interest(verbose=True)

```

Appendix B

Scenario: Cleaning Pipeline Tables

Table B.1: Initial Raw Feature Counts Categorised by Nine Scenarios (Pre Cleaning)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	9	0	7	0	9	0
Numeric with sentinel	187	59	143	60	100	141
Clean numeric	14	27	18	22	20	8
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	248	159	177	220	207	187
Multi class categorical / mixed	267	166	203	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total Columns	809	482	639	592	577	453

Table B.2: Scenario Count Progression: Post Step 1 (Empty / Blank Handling)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	81	31	42	32	75	15
Clean numeric	124	55	123	50	54	134
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.3: Scenario Count Progression: Post Step 2 (Sentinel Standardisation)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	205	86	165	82	129	149
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.4: Scenario Count Progression: Post Step 3 (Numeric Cleaning)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	263	117	215	124	172	149
Censored / bracketed numeric	16	23	17	16	17	73
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	227	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Appendix

Table B.5: Scenario Count Progression: Post Step 4 (Censored / Ordinal Mapping)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	279	140	233	140	189	222
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.6: Scenario Count Progression: Post Step 5 (Binary Classification)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	530	300	413	361	396	409
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.7: Final Dfs Cleaned: Post Step 6 (Multi-Class / Mixed Parsing)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	809	482	639	592	577	453
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	0	0	0	0	0	0
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Appendix C

First Stage: Scenario Variable Mapping

```
1 # Sentinel values (numeric and string)
2 # -1 and -1.0 is sentinel but will be used in the clean pipeline
3 standardised_sentinels = {'-1', '-1.0', -1, -1.0} # set ->
    duplicates removed -> counts correct -> single value detection
    is zero
4 numeric_sentinels = [-1, -1.0, -98, -98.0, -99, -99.0,
5                      95, 95.0, 96, 96.0, 98, 98.0, 99, 99.0]
6 string_sentinels = [str(x) for x in numeric_sentinels] + [f"_{x}"
7                  for x in numeric_sentinels] + \
    ['na', 'n/a', 'refused', 'missing', 'dk', 'rf', 'refused', "
    refuse", 'none', 'no_response', 'nan', 'not_answered', 'no
    answer',
8    'not_applicable', 'no_consent', 'dont_know', "don't_know", "didn
    t_answer", 'prefer_not_to_say', 'task_not_done', '
    insufficient_sample', 'unable_to_record',
9    '__', '*']
10 blanks = ['', ' ', '\t', '\n']
11
12 SCENARIOS = [
13     "Empty_/blank_values",
14     "Numeric_with_sentinel",
15     "Clean_numeric",
16     "Censored_/bracketed_numeric",
17     "Binary_categorical",
```

C. FIRST STAGE: SCENARIO VARIABLE MAPPING

```
18     "Multi_class_categorical/_mixed",
19     "Single_value_categorical",
20     "Special_symbol/_mixed",
21     "Unknown/_complex_mixed"
22 ]
23
24
25 # flatten all dtypes across the waves to float and category
26 def flatten_column_types(df):
27     df_flat = df.copy()
28
29     for col in df_flat.columns:
30         if pd.api.types.is_numeric_dtype(df_flat[col]): # Numeric
31             # columns
32             df_flat[col] = df_flat[col].astype(float) # Convert
33             # to float
34         else:
35             df_flat[col] = ( # Convert to string, strip
36                 whitespace, lowercase, then category
37                 df_flat[col]
38                 .astype(str)
39                 .str.strip()
40                 .str.lower()
41                 .astype('category')
42             )
43
44     return df_flat
45
46 def detect_column_scenario_mapping(df, sample_size=10):
47     summary = []
48
49     # Regex patterns
50     special_symbols_pattern = r'[~A-Za-z0-9\s\.,\-\+&/()]' #
51     # special symbols excluding common numeric formatting
52     bracketed_numeric_pattern = r"\d{1,3}(?:,\d{3})*[-/]\d{1,3}(?:,\d{3})*|\d+\s*[\<>+-]" # patterns like "10-20",
53     # "30/40", ">=50"
54     sentinel_set = set(x.lower() for x in string_sentinels) #
55     # Lowercase string sentinels for comparison
56
57     for col in df.columns:
```

```

52     col_data = df[col].copy()
53     dtype = str(col_data.dtype) # get data type as string
54
55     # Convert to string for sentinel / blank detection
56     str_vals = col_data.astype(str).str.lower().str.strip() #
        lowercase and strip whitespace
57     clean_vals = col_data[~str_vals.isin(sentinel_set) & ~(
        str_vals.isin(blanks))] # remove sentinels and blanks
58
59     # Remove sentinels and blanks for scenario detection only
60     vals_no_sentinel = clean_vals[~clean_vals.astype(str).str
        .lower().isin(sentinel_set)
61                                     & ~clean_vals.astype(str).
        str.lower().isin(blanks)
62                                     ].unique()
63     num_unique_no_sentinel = len(vals_no_sentinel)
64
65     # Check if any sentinel values are present
66     sentinel_present = str_vals.isin(sentinel_set).any()
67
68     # Detection flags
69     empty_detected = str_vals.isin(blanks).any() or len(
        col_data) == 0
70
71     # Numeric detection: coerce to numeric, NaNs for non-
        numeric
72     numeric_vals = pd.to_numeric(clean_vals, errors='coerce')
        # coerce non-numeric to NaN
73     numeric_like = numeric_vals.notna().all() and len(
        clean_vals) > 0 # all values are numeric-like after
        cleaning
74
75     sentinel_detected = str_vals[~str_vals.isin(
        standardised_sentinels)].isin(sentinel_set).any() # or
        col_data.eq(-1.0).any() # check for sentinel presence
76     censored_detected = clean_vals.astype(str).str.contains(
        bracketed_numeric_pattern).any() # check for bracketed
        numeric patterns
77
78     # Binary categorical detection (exactly 2 unique values
        after cleaning)

```

C. FIRST STAGE: SCENARIO VARIABLE MAPPING

```
78     binary_categorical_detected = False
79     if dtype == 'category':
80         if num_unique_no_sentinel == 2:
81             binary_categorical_detected = True
82         elif num_unique_no_sentinel == 1 and sentinel_present
83             :
84             binary_categorical_detected = True
85
86     # Multi-class categorical (3+ unique values, object/
87     # category)
88     multi_class_categorical_detected = num_unique_no_sentinel
89     > 2 and dtype == 'category'
90
91     # Single value categorical
92     single_value_detected = num_unique_no_sentinel == 1 and
93     not binary_categorical_detected
94
95     # Special symbols detection (ignore numeric formatting)
96     symbol_detected = clean_vals.astype(str).str.contains(
97     special_symbols_pattern).any()
98
99     # Mixed types detection
100    mixed_detected = len(set(type(v) for v in col_data)) > 1
101    # more than one data type present
102
103    # Determine scenario in order of pipeline
104    if empty_detected:
105        scenario = SCENARIOS[0]
106        handling = "Replace_numeric/categorical_empty/blank_
107        with_-1.0"
108    elif numeric_like and sentinel_detected:
109        scenario = SCENARIOS[1]
110        handling = "Replace_numeric_sentinel_values_with_-1.0
111        "
112    elif numeric_like and not sentinel_detected:
113        scenario = SCENARIOS[2]
114        handling = "Keep_as_numeric"
115    elif censored_detected:
116        scenario = SCENARIOS[3]
117        handling = "Convert_numeric_ranges_to_midpoints_or_
118        keep_as_categorical"
```

```

110     elif binary_categorical_detected:
111         scenario = SCENARIOS[4]
112         handling = "Encode as 0/1 or keep as category"
113     elif multi_class_categorical_detected:
114         scenario = SCENARIOS[5]
115         handling = "One-hot encode, label encode, or keep as
116                     category"
117     elif single_value_detected:
118         scenario = SCENARIOS[6]
119         handling = "Keep as category / constant column"
120     elif symbol_detected:
121         scenario = SCENARIOS[7]
122         handling = "Inspect and preprocess based on context"
123     else:
124         scenario = SCENARIOS[8]
125         handling = "Inspect manually / complex mixed types"
126
127     sample_values = list(col_data.unique()[:sample_size]) #
128                     sample unique values
129
130     summary.append({
131         'Variable': col,
132         'DataType': dtype,
133         'UniqueValues': num_unique_no_sentinel,
134         'Scenario': scenario,
135         'SuggestedHandling': handling,
136         'SampleValues': sample_values
137     })
138
139     return pd.DataFrame(summary)

```

Appendix D

Step 6: Scenario ‘Multi class categorical’ Cleaning Function

```
1 def categorical_transform_S6(df): # Step6; Scenario -> Multi
   class categorical
2     df_cleaned = df.copy()
3     sent_set = set([str(x).lower() for x in string_sentinels]) |
        set([str(x).lower() for x in standardised_sentinels])
4     numeric_sentinel_set = set(float(x) for x in
        numeric_sentinels)
5
6     # Helper normaliser
7     def _norm(tok):
8         if tok is None:
9             return None
10        s = str(tok).strip().lower()
11        if s in blanks:
12            return ''
13        return s
14
15    # Words -> number map
16    word_num_map = {
17        'zero':0.0, 'one':1.0, 'two':2.0, 'three':3.0, 'four':4.0, '
        five':5.0, 'six':6.0, 'seven':7.0, 'eight':8.0, 'nine'
        :9.0,
18        'ten':10.0, 'eleven':11.0, 'twelve':12.0, 'thirteen':13.0, '
        fourteen':14.0, 'fifteen':15.0, 'twenty':20.0,
```

```

19         'thirty':30.0, 'forty':40.0, 'fifty':50.0, 'sixty':60.0, '
20         seventy':70.0, 'eighty':80.0, 'ninety':90.0,
21         'a_few':2.0, 'a_couple':2.0, 'few':2.0, 'once':1.0, 'twice'
22         :2.0, 'several':3.0, 'couple':2.0
23     }
24
25     def words_to_num(token):
26         if token is None:
27             return None
28         token = token.strip().lower()
29         if token in word_num_map:
30             return word_num_map[token]
31         try:
32             return float(token)
33         except Exception:
34             return None
35
36     # Regexes for parsing
37     RE_TIME = re.compile(r'^s*(\d{1,2}):(\d{2})\s*$')
38     RE_PERCENT = re.compile(r'^s*(-?\d+(?:\.\d+)?)\s%\s*$')
39     RE_DIRECT_NUM = re.compile(r'^s*-?\d+(?:\.\d+)?\s*$')
40     RE_RANGE = re.compile(r'^s*(-?\d+(?:\.\d+)?)\s*[-\s*/]\s*(-?\d+(?:\.\d+)?)\s*$')
41     RE_RANGE_TO = re.compile(r'^s*(-?\d+(?:\.\d+)?)\s*(?:to|TO)\s*(-?\d+(?:\.\d+)?)\s*$')
42     RE_PLUS = re.compile(r'^s*(-?\d+(?:\.\d+)?)\s*\+\s*$')
43     RE_GT = re.compile(r'^s*(?:>|=|>)\s*(-?\d+(?:\.\d+)?)\s*$')
44     RE_LT_ANCHOR = re.compile(r'^s*(?:<|<=)\s*(-?\d+(?:\.\d+)?)')
45     RE_NUM_FIND = re.compile(r'-?\d+(?:\.\d+)?')
46     RE_WORD_TOK = re.compile(r"[a-z']+(?:\s[a-z']+)?")
47     # RE_SURVEY_TIME = re.compile(r'^s*(?:[a-z]+\s+)?(\d+(?:\.\d+)+?)\s*(?:month|week|year)s?.*$', re.IGNORECASE)
48
49     def norm_return(v):
50         try:
51             vf = float(v)
52         except Exception:
53             return None
54         if vf in numeric_sentinel_set:
55             return 'SENTINEL'
56         return vf

```

D. STEP 6: SCENARIO ‘MULTI CLASS CATEGORICAL’ CLEANING FUNCTION

```
55 def try_parse_token(x):
56     """Return float, SENTINEL, or None for unknown token x"""
57     if x is None:
58         return None
59
60     # preserve numeric pre transformed values
61     if isinstance(x, (int, float)):
62         if float(x) in numeric_sentinel_set:
63             return 'SENTINEL'
64         return float(x)
65
66     s = _norm(x)
67     # Extending sentinel detection for categorical parsing
68     # scenario
69     if s == '':
70         return 'SENTINEL'
71     if 'not_applicable' in s or 'does_not_have' in s or "don't
72         receive" in s or "don't receive" in s or 'unknown'
73         in s:
74         return 'SENTINEL'
75     if s in sent_set or re.fullmatch(r'[0-9a-z]+', s):
76         return 'SENTINEL'
77     if s in ('other', 'other_specify') or s.startswith('other
78         ') or s.endswith('(specify)':
79         return 'SENTINEL'
80
81     s0 = s.replace(',', '').replace('$', '').replace(EUR, '').
82         replace('#', '')
83
84     s0 = re.sub(r'(\d+)-', r'\1_', s0)
85     s0 = re.sub(r'-(\d+)', r'_\1', s0)
86
87     RE_PREFIX_NUM = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\.\s*.\s*
88         $')
89
90     m = RE_PREFIX_NUM.match(s0)
91     if m:
92         # If the format is 'NUM. anything', return the number
93         # as the category value
94         return norm_return(float(m.group(1)))
```

```

88
89     # time
90     m = RE_TIME.match(s0)
91     if m:
92         h, mi = float(m.group(1)), float(m.group(2))
93         return norm_return(h * 60.0 + mi)
94
95     # percent
96     m = RE_PERCENT.match(s0)
97     if m:
98         return norm_return(float(m.group(1)))
99
100    # direct numeric
101    if RE_DIRECT_NUM.match(s0):
102        return norm_return(s0)
103
104    # ranges
105    m = RE_RANGE.match(s0)
106    if m:
107        try:
108            a, b = float(m.group(1)), float(m.group(2))
109            return norm_return((a + b) / 2.0)
110        except Exception:
111            pass
112    m = RE_RANGE_T0.match(s0)
113    if m:
114        try:
115            a, b = float(m.group(1)), float(m.group(2))
116            return norm_return((a + b) / 2.0)
117        except Exception:
118            pass
119
120    # plus / greater than
121    m = RE_PLUS.match(s0) or RE_GT.match(s0)
122    if m:
123        nums = RE_NUM_FIND.findall(s0)
124        if nums:
125            return norm_return(nums[0])
126
127    # less than
128    if RE_LT_ANCHOR.search(s0) or 'less_than' in s0:
```

D. STEP 6: SCENARIO ‘MULTI CLASS CATEGORICAL’ CLEANING FUNCTION

```
129         nums = RE_NUM_FIND.findall(s0)
130         if nums:
131             return norm_return(nums[0])
132
133     # or less / more
134     m = re.search(r'(-?\d+(?:\.\d+)?)\s*or\s*less', s0)
135     if m:
136         return norm_return(m.group(1))
137     m = re.search(r'(-?\d+(?:\.\d+)?)\s*or\s*more', s0)
138     if m:
139         return norm_return(m.group(1))
140
141     # "more than X but less than Y", "over X but less than Y"
142     m = re.search(r'(?:(more|than|over)\s*e?(-?\d+(?:\.\d+)?)\s*
143                  s*(?:but\s*)?(?:less|than|under)\s*e?(-?\d+(?:\.\d+)?)
144                  ', s0)
145     if m:
146         try:
147             a, b = float(m.group(1)), float(m.group(2))
148             return norm_return((a + b) / 2.0)
149         except Exception:
150             pass
151
152     # "between X and Y" / "between X - Y"
153     m = re.search(r'between\s*e?(-?\d+(?:\.\d+)?)\s*(?:and|[-
154                  --/])\s*e?(-?\d+(?:\.\d+)?)', s0)
155     if m:
156         try:
157             a, b = float(m.group(1)), float(m.group(2))
158             return norm_return((a + b) / 2.0)
159         except Exception:
160             pass
161
162     # survey time based
163     time_map_weeks = {
164         'one|week': 1.0,
165         'two|weeks': 2.0,
166         'a|month|(4|weeks)': 4.0,
167         'a|month/4|weeks': 4.0,
168         'three|months|(13|weeks)': 13.0,
169         'three|months/13|weeks': 13.0,
```



```

167         'six_months_(26_weeks)': 26.0,
168         'six_months/26_weeks': 26.0,
169         'one_year_(12_months/52_weeks)': 52.0,
170         'one_year/12_months/52_weeks': 52.0
171     }
172     if s0 in time_map_weeks:
173         return norm_return(time_map_weeks[s0])
174
175     # If a category is '3 months or more', try to extract the
176     # number and use upper bound
177     if 'months_or_more' in s0 and RE_NUM_FIND.search(s0):
178         nums = RE_NUM_FIND.findall(s0)
179         if nums:
180             months = float(nums[0])
181             return norm_return(months * 4.33) # Convert
182             months to weeks (4.33 weeks per month)
183
184     # Frequency calendar based
185     if any(tok in s0 for tok in ('day', 'week', 'month', 'year',
186         'daily', 'never', 'almost_every_day', 'once', 'twice',
187         'thrice', 'three', 'several', 'couple')):
188         nums = RE_NUM_FIND.findall(s0)
189         if nums:
190             val = (float(nums[0]) + float(nums[1])) / 2.0 if
191                 len(nums) >= 2 else float(nums[0])
192         else:
193             if 'once_or_twice' in s0:
194                 val = 1.5
195             else:
196                 val = None
197                 for w in RE_WORD_TOK.findall(s0):
198                     wn = words_to_num(w)
199                     if wn is not None:
200                         val = wn
201                         break
202     if val is not None:
203         if 'day' in s0:
204             if 'week' in s0:
205                 return norm_return(val * 4.33)
206             if 'month' in s0:
207                 return norm_return(val)

```

D. STEP 6: SCENARIO 'MULTI CLASS CATEGORICAL' CLEANING FUNCTION

```
203         if 'daily' in s0 or 'every_day' in s0 or '
204             almost_every_day' in s0:
205             return norm_return(val * 30.0)
206             return norm_return(val * 4.33)
207         if 'week' in s0:
208             return norm_return(val * 4.33)
209         if 'month' in s0:
210             return norm_return(val)
211         if 'year' in s0:
212             return norm_return(val / 12.0)
213         return norm_return(val)
214
215     likert_map = {
216         # Likert / ordinals (expanded with quintile/lowest/
217             highest detection)
218         'strongly_disagree': 0.0, 'disagree': 1.0, 'neither_
219             agree_nor_disagree': 2.0, 'neither': 2.0,
220         'agree': 3.0, 'strongly_agree': 4.0,
221         'very_poor': 0.0, 'poor': 1.0, 'fair': 2.0, 'good':
222             3.0, 'very_good': 4.0, 'excellent': 5.0,
223         'none': 0.0, 'not_at_all': 0.0, 'hardly_ever': 0.0,
224         'a_little': 1.0, 'some': 2.0, 'a_lot': 3.0, 'alot':
225             3.0, 'often': 3.0,
226         'yes': 1.0, 'no': 0.0, 'sometimes': 2.0, 'rarely':
227             1.0, 'most_or_all_of_the_time': 4.0,
228         'pass': 2.0, 'fail': 1.0, '100': 100.0,
229         # Problem severity mappings
230         'no_problem': 1.0, 'minor_problem': 2.0, 'moderate_
231             problem': 3.0, 'major_problem': 4.0,
232         # Concern level mappings
233         'not_at_all_concerned': 1.0, 'somewhat_concerned':
234             2.0, 'fairly_concerned': 3.0, 'very_concerned': 4.0,
235         # Survey frequency mappings
236         'daily/_almost_daily': 7.0, 'once_a_week_or_more':
237             6.0, 'twice_a_month_or_more': 5.0, 'about_once_a_
238             month': 4.0,
239         'every_few_months': 3.0, 'about_once_or_twice_a_year'
240             : 2.0, 'less_than_once_a_year': 1.0, 'never': 0.0,
241         'daily': 7.0, 'every_day': 7.0, 'almost_every_day':
242             7.0, 'most_or_all_of_the_time': 7.0,
```

```

232         'every_week': 6.0, 'once_a_week': 6.0, 'several_times_
           per_week': 6.0,
233         'twice_a_month': 5.0, 'two_a_month': 5.0, 'several_
           times_per_month': 5.0, 'once_or_twice_a_month':
           5.0,
234         'once_a_month': 4.0, 'once_or_twice_a_year': 2.0, '
           less_than_once_a_month': 1.0,
235     }
236
237     if s0 in likert_map:
238         return float(likert_map[s0])
239
240     # Quintile / ordinal like tokens: 'lowest', '2nd quintile
           ', 'highest'
241     if 'quintile' in s0 or 'lowest' in s0 or 'highest' in s0
           or 'middle' in s0 or 'median' in s0:
242         if 'lowest' in s0:
243             return norm_return(1.0)
244         if 'highest' in s0:
245             return norm_return(5.0)
246         if 'middle' in s0 or 'median' in s0:
247             return norm_return(3.0)
248         mq = re.search(r'(\d+)(?:st|nd|rd|th)?\s+quintile
           ', s0)
249         if mq:
250             return norm_return(float(mq.group(1)))
251         nums = RE_NUM_FIND.findall(s0)
252         if nums:
253             return norm_return(nums[0])
254
255     if 'adjacent' in s0: # Adjacent boxes: keep semantic
           average if present
256         nums = RE_NUM_FIND.findall(s0)
257         if len(nums) >= 2:
258             return norm_return((float(nums[0]) + float(nums
           [1])) / 2.0)
259         elif nums:
260             return norm_return(nums[0])
261         return None
262
263     # Fallback: first number

```

D. STEP 6: SCENARIO ‘MULTI CLASS CATEGORICAL’ CLEANING FUNCTION

```
264     nums = RE_NUM_FIND.findall(s0)
265     if nums:
266         return norm_return(nums[0])
267
268     return None
269
270 def category_sort_key(u):
271     us = _norm(u) or ''
272     parsed = try_parse_token(us)
273     if isinstance(parsed, (int, float)):
274         return (0, float(parsed), us)
275     if us == 'no' or us.startswith('no_') or us.startswith('
276         not_'):
277         return (1, 0.0, us)
278     if 'yes' in us:
279         if 'both' in us or 'two' in us:
280             return (1, 2.0, us)
281         if 'one' in us:
282             return (1, 1.0, us)
283         return (1, 1.0, us)
284     amount_keywords = ['none', 'not_at_all', 'hardly_ever', '
285         a_little', 'some', 'a_lot', 'alot', 'often', '
286         sometimes', 'rarely']
287     for i, key in enumerate(amount_keywords):
288         if key in us:
289             return (2, float(i), us)
290     if 'adjacent_boxes' in us or 'adjacent_numbers' in us:
291         nums = RE_NUM_FIND.findall(us)
292         if len(nums) >= 2:
293             a, b = float(nums[0]), float(nums[1])
294             return (10, (a + b) / 2.0, us)
295         elif nums:
296             return (10, float(nums[0]), us)
297         return (10, 0, us)
298     return (99, 0, us)
299
300 # Operate only on categorical columns
301 cat_cols = df_cleaned.select_dtypes(include='category').
302     columns.tolist()
303 for col in cat_cols:
```

```

300     uniques = list(dict.fromkeys(df_cleaned[col].tolist())) #
           Preserve original tokens with types (do NOT cast to
           str here)
301     normalised_pairs = [(u, _norm(u)) for u in uniques] #
           Build normalised pairs keeping original token too
302     # Keep tokens that are non sentinel and parseable (
           strings or numbers)
303     non_sentinel = [u for u, nu in normalised_pairs if (nu
           and nu not in sent_set and nu not in blanks and
           try_parse_token(nu) != 'SENTINEL')]
304
305     if not non_sentinel:
306         continue
307
308     # Computing parse results keyed by normalized string (but
           keep list of originals too)
309     parse_results = { _norm(u): try_parse_token(_norm(u)) for
           u in non_sentinel }
310
311     numeric_count = sum(1 for v in parse_results.values() if
           isinstance(v, (int, float)))
312     sentinel_parsed_count = sum(1 for v in parse_results.
           values() if v == 'SENTINEL')
313     all_parse_numeric = (numeric_count +
           sentinel_parsed_count == len(parse_results))
314
315     if all_parse_numeric:
316         # Map token string -> numeric (SENTINEL -> -1.0)
317         code_map = { k: (float(v) if isinstance(v, (int,
           float)) else -1.0) for k, v in parse_results.items
           () }
318
319     def map_to_numeric(x):
320         # Preserve the already numeric cells (except
           numeric sentinels)
321         if isinstance(x, (int, float)) and not pd.isna(x)
           :
322             if float(x) in numeric_sentinel_set:
323                 return -1.0
324             return float(x)
325         if pd.isna(x):

```

D. STEP 6: SCENARIO ‘MULTI CLASS CATEGORICAL’ CLEANING FUNCTION

```
326         return -1.0
327     xs = _norm(x)
328     if not xs or xs in sent_set or xs in blanks or re
329         .fullmatch(r'[~0-9a-z]+', xs):
330         return -1.0
331     return float(code_map.get(xs, -1.0))
332
333 df_cleaned[col] = df_cleaned[col].apply(
334     map_to_numeric).astype(float)
335
336 else:
337     first_seen_norm = []
338     seen = set()
339     for u in non_sentinel:
340         nu = _norm(u)
341         if nu not in seen:
342             first_seen_norm.append(nu)
343             seen.add(nu)
344
345 parse_results = {nu: try_parse_token(nu) for nu in
346     first_seen_norm}
347 max_reserved_val = -1.0
348 for nu, val in parse_results.items():
349     if isinstance(val, (int, float)):
350         max_reserved_val = max(max_reserved_val, val)
351         # Assuming any parsed float is a reserved
352         # ID if it came from the map
353 start_i = int(max_reserved_val) + 1
354 if start_i == 0: # Ensure we never start at 0 if 0 is
355     a reserved category i.e for ('no')
356     start_i = 1
357 string_tokens = [nu for nu in first_seen_norm if not
358     isinstance(parse_results.get(nu), (int, float))]
359 sorted_strings = sorted(string_tokens, key=
360     category_sort_key)
361
362 code_map_strings = { norm_tok: i for i, norm_tok in
363     enumerate(sorted_strings, start=start_i) }
364
365 def map_mixed(x):
```

```

357         if isinstance(x, (int, float)) and not pd.isna(x)
           :# Preserve real numeric cell values
358             if float(x) in numeric_sentinel_set:
359                 return -1.0
360             return float(x)
361         if pd.isna(x):
362             return -1.0
363         xs = _norm(x)
364         if xs == '' or xs in sent_set or xs in blanks or
           re.fullmatch(r'[^0-9a-z]+', xs):
365             return -1.0
366
367         parsed_val = try_parse_token(xs)
368         if parsed_val == 'SENTINEL':
369             return -1.0
370         if isinstance(parsed_val, (int, float)):#
           Captures 'no' (0.0), 'yes' (1.0), and other
           hardcoded likert values.
371             return float(parsed_val)
372         if xs in code_map_strings:
373             return float(code_map_strings[xs])
374         return -1.0
375
376         df_cleaned[col] = df_cleaned[col].apply(map_mixed).
           astype(float)
377
378     # Final sweep on numeric columns
379     for c in df_cleaned.select_dtypes(include='float').columns:
380         df_cleaned[c] = df_cleaned[c].fillna(-1.0)
381         df_cleaned[c] = df_cleaned[c].replace(list(
           numeric_sentinel_set), -1.0)
382
383     return df_cleaned

```

Appendix E

Second Stage: Deterministic *six* *step* Cleaning Pipeline

```
1 # Cleaning pipeline to all waves
2 dfs_cleaned = {}
3
4 # flattened dtype df
5 dfs_flat = {wave: flatten_column_types(df) for wave, df in dfs.
6             items()} # flatten dtypes
7
8 # Original df
9 display(scenario_count_table(dfs_flat).style.set_caption("
10     Original Table - pre cleaning")) # display original scenario
11     count table
12
13 # Step 1: Clean empty/missing values
14 dfs_step1 = {wave: clean_empty_missing(df) for wave, df in
15             dfs_flat.items()} # apply step 1 cleaning
16 display(scenario_count_table(dfs_step1).style.hide(axis="index").
17         set_caption("Step 1: Empty / Blank - post cleaning"))
18
19 # Step 2: Clean sentinel values
20 dfs_step2 = {wave: clean_sentinel(df) for wave, df in dfs_step1.
21             items()} # apply step 2 cleaning
22 display(scenario_count_table(dfs_step2).style.hide(axis="index").
23         set_caption("Step 2: Sentinel - post cleaning"))
24
```

```

18 # Step 3: Numerical Transformation
19 dfs_step3 = {wave: numerical_transform(df) for wave, df in
    dfs_step2.items()} # apply step 3 cleaning
20 display(scenario_count_table(dfs_step3).style.hide(axis="index").
    set_caption("Step 3: Numeric Transformation - post cleaning"))
21
22 # Step 4: Categorical Transformation for censored scenario
23 dfs_step4 = {wave: categorical_transform_S4(df) for wave, df in
    dfs_step3.items()} # apply step 4 cleaning
24 display(scenario_count_table(dfs_step4).style.hide(axis="index").
    set_caption("Step 4: Category Transformation S4 - post
    cleaning"))
25
26 # Step 5: Categorical Transformation for binary values
27 dfs_step5 = {wave: categorical_transform_S5(df) for wave, df in
    dfs_step4.items()} # apply step 5 cleaning
28 display(scenario_count_table(dfs_step5).style.hide(axis="index").
    set_caption("Step 5: Category Transformation S5 - post
    cleaning"))
29
30 # Step 6: Categorical Transformation for multi class/mixed values
31 dfs_step6 = {wave: categorical_transform_S6(df) for wave, df in
    dfs_step5.items()} # apply step 6 cleaning
32 display(scenario_count_table(dfs_step6).style.hide(axis="index").
    set_caption("Step 6: Category Transformation S6 - post
    cleaning"))
33
34 dfs_cleaned = dfs_step6

```

Appendix F

MD Dictionary: Scenario Grouping and Transformation Function

```
1 def show_grouped_vars_for_scenarios(dfs_original, dfs_transformed
  , scenarios_to_show): # Function to show grouped variables per
  scenario across waves -> markdown format
2     """
3     Used to inspect the scenario variables/features that need
  handling for transformation/cleaning.
4     A pre and post cleaning helper function to identify variables
  that need specific handling and to
5     Inspect actual changes for a scenario from previous cleaning
  step to current step.
6     """
7
8     print("# Category Transformation Mapping")
9     # print(f"\n## Scenarios: {scenarios_to_show}") # Markdown
  mapping dictionary header
10    for wave in dfs_original.keys():
11        print(f"\n### ===== {wave.upper()} =====")
12
13        mapping_df = detect_column_scenario_mapping(
            dfs_transformed[wave]) # Get scenario mapping from
            transformed/cleaned to see overall changes for all
            variables
```

```

14     # mapping_df = detect_column_scenario_mapping(
15         dfs_original[wave]) # Get scenario mapping from
16         original to see untransformed values
17
18     for scenario in scenarios_to_show:
19         vars_in_scenario = mapping_df[mapping_df['Scenario']
20             == scenario]['Variable'].tolist() # Variables in
21             scenario
22         print(f"\n---_Scenario_{scenario}:_{len(
23             vars_in_scenario)}_---") # Scenario header for
24             pipeline git commit
25         if not vars_in_scenario:
26             print("\nNo_variables_found.")
27             continue
28
29         # Dictionary: frozen set(values) -> list of variables
30         groups = {}
31
32         for var in vars_in_scenario:
33             unique_vals = list(dict.fromkeys(dfs_original[
34                 wave][var].dropna())) # Preserve first seen
35                 order from ORIGINAL
36             key = tuple(unique_vals) # Use ordered tuple as
37                 key
38
39             if key not in groups: # Initialise list if key
40                 not present
41                 groups[key] = []
42             groups[key].append(var) # Append variable to the
43                 group
44
45         # Print grouped variables in markdown format friendly
46         for Notion -> mapping dictionary
47         for value_set, var_list in groups.items():
48             print("\n*Variables:*","" + ",".join(var_list)
49                 + "'_") # \n
50             print(f"*Pre_transformed_values:*_{list(value_set
51                 )}_")
52
53         # Pull transformed values from dfs_cleaned /
54         specific step using preserved order

```

Appendix

```
40         try:
41             df_before = dfs_original[wave][var_list[0]]
42             df_after = dfs_transformed[wave][var_list[0]]
43
44             # Build transformed list in same order as
45             # original values
46             transformed_vals = []
47             for orig_val in value_set: # Iterate through
48                 # original values in order
49                 mask = df_before == orig_val
50                 if mask.any():
51                     trans_vals = df_after[mask].
52                     unique().tolist() # Get unique
53                     transformed values
54                 if trans_vals:
55                     transformed_vals.append(
56                         trans_vals[0]) # Append
57                     first transformed value
58
59         except Exception:
60             transformed_vals = []
61         print(f"*Transformed_numeric:*_{transformed_vals}
62             {")")
63
64 scenarios_to_show = SCENARIOS
65 # scenarios_to_show = ["Clean numeric"]
66 # scenarios_to_show = ["Censored / bracketed numeric"]
67 # scenarios_to_show = ["Binary categorical"]
68 # scenarios_to_show = ["Multi class categorical / mixed"]
69
70 show_grouped_vars_for_scenarios(dfs, dfs_cleaned,
71     scenarios_to_show) # prev step == comparison to current step
72     else current step transformed/cleaned for ready .md dictionary
```

Appendix G

MD Dictionary: Scenario Grouping and Transformation Audit Outputs

- **sex**: Binary biological sex indicator (e.g., **Male/Female**, **Female/Male**, pre-fixed encodings).
- **ph001**: Self-rated general health status measured on an ordinal scale. Survey question: *“Now I would like to ask you some questions about your health. Would you say your health is. . .”*. Responses range from *Poor* to *Excellent*, with additional ‘no response’ categories.
- **ph009**: Comparative self-rated health relative to people of the same age. Survey question: *“In general, compared to other people your age, would you say your health is. . .”*.
- **bh006**: Frequency based lifestyle variable combining numeric values, bounded ranges (e.g., 20-39), and textual descriptors (e.g., **Less than 10**, 30+). Survey question: *“How many cigarettes do/did you smoke on average per day?”*.

```
%-----%
%-----MARKDOWN_MAPPING_DICTIONARY-----%
%-----%
# Category Transformation Mapping
```

G. MD DICTIONARY: SCENARIO GROUPING AND TRANSFORMATION AUDIT OUTPUTS

===== WAVE1 =====

--- Scenario Empty / blank values: 0 ---

No variables found.

--- Scenario Numeric with sentinel: 0 ---

No variables found.

--- Scenario Clean numeric: 809 ---

Variables: 'sex, gd002'

Pre transformed values: ['Male', 'Female']

Transformed numeric: '[1.0, 0.0]'

Variables: 'ph001'

Pre transformed values: ['Fair', 'Good', 'Very good', 'Excellent', 'Poor', "Don't k

Transformed numeric: '[2.0, 3.0, 4.0, 5.0, 1.0, -1.0]'

Variables: 'bh006'

Pre transformed values: ['20-39', 10.0, -1.0, 15.0, 5.0, '60+', "Don't Know", 2.0, 8

Transformed numeric: '[29.5, 10.0, -1.0, 15.0, 5.0, 60.0, -1.0, 2.0, 8.0, 3.0, 49.5

--- Scenario Censored / bracketed numeric: 0 ---

No variables found.

--- Scenario Binary categorical: 0 ---

No variables found.

--- Scenario Multi class categorical / mixed: 0 ---

No variables found.

--- Scenario Single value categorical: 0 ---

No variables found.

--- Scenario Special symbol / mixed: 0 ---

No variables found.

--- Scenario Unknown / complex mixed: 0 ---

No variables found.

===== WAVE2 =====

...

--- Scenario Clean numeric: 482 ---

Variables: 'gd002, sex'

Pre transformed values: ['Female', 'Male']

Transformed numeric: '[0.0, 1.0]'

Variables: 'bh006'

Pre transformed values: [-1.0, '10 - 19', '20 - 29', 'Less than 10', '30+',

Transformed numeric: '[-1.0, 14.5, 24.5, 10.0, 30.0, -1.0]'

Variables: 'ph001'

Pre transformed values: ['Very good', 'Good', 'Fair', 'Excellent', 'Poor']

Transformed numeric: '[4.0, 3.0, 2.0, 5.0, 1.0]'

...

G. MD DICTIONARY: SCENARIO GROUPING AND TRANSFORMATION AUDIT OUTPUTS

===== WAVE3 =====

...

--- Scenario Clean numeric: 639 ---

Variables: 'sex, gd002'

Pre transformed values: ['Female', 'Male']

Transformed numeric: '[0.0, 1.0]'

Variables: 'ph001, ph009'

Pre transformed values: ['Very Good', 'Good', 'Excellent', 'Fair', 'Poor', 'DK']

Transformed numeric: '[4.0, 3.0, 5.0, 2.0, 1.0, -1.0]'

Variables: 'bh006'

Pre transformed values: [-1.0, '10 - 19', 'Less than 10', '20 - 29', '30+', -98.0,

Transformed numeric: '[-1.0, 14.5, 10.0, 24.5, 30.0, -1.0, -1.0]'

...

===== WAVE4 =====

...

--- Scenario Clean numeric: 592 ---

Variables: 'bh006'

Pre transformed values: [-1.0, 'Less than 10', '20 - 29', '10 - 19', '30+', "Don't

Transformed numeric: '[-1.0, 10.0, 24.5, 14.5, 30.0, -1.0]'

Variables: 'gd002, sex'

Pre transformed values: ['Female', 'Male']

Transformed numeric: '[0.0, 1.0]'

```

*Variables:* 'ph001'
*Pre transformed values:* ['Good', 'Fair', 'Poor', 'Excellent', 'Very Good', 'I
*Transformed numeric:* '[3.0, 2.0, 1.0, 5.0, 4.0, -1.0, -1.0]'

...

### ===== WAVE5 =====

...

--- Scenario Clean numeric: 577 ---

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, 'Less than 10', '20 - 29', '10 - 19', '30+',
*Transformed numeric:* '[-1.0, 10.0, 24.5, 14.5, 30.0, -1.0]'

*Variables:* 'gd002, sex'
*Pre transformed values:* ['Female', 'Male']
*Transformed numeric:* '[0.0, 1.0]'

*Variables:* 'ph001'
*Pre transformed values:* ['Very Good', 'Fair', 'Excellent', 'Good', 'Poor']
*Transformed numeric:* '[4.0, 2.0, 5.0, 3.0, 1.0]'

...

```

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